# 51 The Pathophysiology of Uremia

Sahir Kalim | S. Ananth Karumanchi | Anders H. Berg

## CHAPTER OUTLINE

SOLUTES CLEARED BY THE KIDNEY AND RETAINED IN UREMIA, 1515
SOLUTE REMOVAL BY DIFFERENT FORMS OF KIDNEY REPLACEMENT THERAPY, 1523
METABOLIC EFFECTS OF UREMIA, 1526
CONCLUSIONS, 1530

## KEY POINTS

- Uremia is a clinical syndrome with a wide range of signs and symptoms rather than a well-circumscribed disease.
- The uremic syndrome is likely caused by the accumulation of organic chemicals normally cleared by the kidneys.
- Knowledge of which specific solutes cause uremia remains hampered by the broad array of signs and symptoms associated with uremia and the large number of known and unknown retention solutes that need to be studied to definitively ascribe a mechanistic link.
- Uremia is only partially relieved through enhanced solute clearance via dialysis, highlighting complexities in solute clearance and the limitations of traditional dialysis itself.
- Certain compounds, including the prototypical retention solute urea, can permanently modify proteins (e.g., protein carbamylation) and alter molecular signaling, creating cascading "downstream" effects further complicating the therapeutic targeting of the most significant uremic toxins.

Kidney disease disrupts two important functions of the kidney: 1. clearance of water and renally excreted waste products and 2. synthesis of products such as erythropoietin. "Uremia" is a broad term originally coined by Pierre Adolph Piorry because of its association with elevated concentrations of urea in the blood,[1] but modern definitions of uremia are variable and usually detail a diffuse clinical syndrome encompassing a range of signs and symptoms, rather than a well-defined disease entity.[2] Indeed, authors in prior editions of this book succinctly described uremia as the ill effects of kidney failure that we cannot yet explain. Hypertension caused by volume overload and upregulation of the renin-angiotensin-aldosterone system, tetany caused by hypocalcemia, and anemia caused by erythropoietin deficiency were once considered uremic signs but were recategorized as their causes were discovered. Uremia may thus now be defined as the signs, symptoms, and sequelae caused by accumulation of renally excreted toxic metabolites that are independent of changes in extracellular volume and inorganic ion concentrations and independent of the loss of physiologically important known (and undiscovered) renal synthetic products (Box 51.1). Despite tremendous efforts, our knowledge of which specific solutes drive uremia remains limited. Newer and updated conceptualization of uremia acknowledges two additional points. First, despite the development and advancement of solute clearance through dialysis, while some uremic symptoms improve readily, other symptoms persist, implying that dialysis clearance is insufficient to completely treat uremia. Second, certain solutes including urea can have indirect effects on important proteins and molecular signaling, creating cascading "downstream" uremic effects that make it even more difficult to precisely identify and study the most significant uremic toxins.[3]

Because uremia arises with advanced kidney failure, it is commonly attributed to the accumulation of metabolic waste products and organic chemicals that would otherwise be cleared by healthy kidneys (Fig. 51.1). In general, the study of renal organic waste removal lags far behind the study of inorganic ion excretion. A major problem is the multiplicity of waste solutes to study. The first comprehensive review prepared by the European Uremic Toxin Work Group in 2003 (EUTox[4,5]) listed >80 uremic solutes, and further studies have increased this number to >250.[6-10] Untargeted mass spectrometry has revealed that the true total number of metabolites that accumulate with renal insufficiency is even greater and that the majority of uremic solutes cannot be found in standard databases of known biological

**Box 51.1 Metabolic Effects, Symptoms, and Signs of Uremia**

**Metabolic**

Increased oxidant levels
Reduced resting energy expenditure
Reduced body temperature
Insulin resistance
Muscle wasting
Amenorrhea and sexual dysfunction

**Neural and Muscular**

Fatigue
Altered mental status ranging from loss of concentration to coma and seizures
Sleep disturbances
Restless legs
Peripheral neuropathy
Anorexia and nausea
Diminution in taste and smell
Itching
Cramps
Reduced muscle membrane potential

**Other**

Serositis (including pericarditis)
Hiccups
Granulocyte and lymphocyte dysfunction
Platelet dysfunction
Shortened erythrocyte life span
Albumin modifications

compounds. With so many substances to study, it is hard to establish which ones are toxic. Bergstrom[11] suggested criteria for identifying uremic toxins that are analogous to Koch's postulates for identifying infectious agents, and others have updated these to reflect limitations associated with the original terminology.[10,12] According to these criteria, a uremic toxin must have a known chemical structure or be able to be identified and quantified accurately in the blood, plasma, or serum, along with the following features:

- Its concentrations in a biospecimen should be higher in patients with kidney disease than in people with normal kidney function.
- The solute should demonstrate negative effects or contribute to specific uremic symptoms in vivo, ex vivo, or in vitro.
- Biologically active concentrations of the solute in studies should conform to those observed in patients with chronic kidney disease.

Given the complexity of the uremic milieu, it is unlikely that the accumulation of a single solute in isolation could recapitulate uremia or that removal of the same could eliminate uremia, thus complicating and slowing progress in uremia research. Nevertheless, our knowledge of solutes that build up with uremia has increased significantly in recent years, prompting expert panels to at least identify limitations in the definitions and classifications of uremic retention solutes and toxins, as well as rank the evidence score for the various toxins that have been studied.[10]

Most solutes that accumulate with advanced chronic kidney disease (CKD) are probably not toxic. Others that are toxic may exert their ill effects only when present in combination. The difficulty imposed by the multiplicity of solutes is compounded by the multiplicity of signs and symptoms encountered in patients diagnosed with uremia. Investigators of uremic toxicity thus face the daunting task of matching a solute or group of solutes to an appropriate endpoint. Many of the effects of uremia are also hard to quantify, which makes the problem even more difficult. This is particularly true of common major uremic symptoms such as fatigue, anorexia, nausea, pruritis, paresthesia, pain, and diminished mental acuity. A related problem encountered in studies of uremia is distinguishing the effects of uremia from those of related conditions. Paradoxically, the widespread availability of dialysis has made uremia even more difficult to study. The severity of the classic uremic symptoms may be attenuated with the commencement of dialysis. Yet patients now suffer from a new illness, which Depner[13] has named the "residual syndrome," composed of partially treated uremia, the side effects of the dialysis procedure, and the residual risk of sequelae despite repeated correction of fluid and electrolyte imbalances. In most patients, features of the residual syndrome are further combined with the effects of age and of systemic diseases responsible for their kidney failure. Disturbances of inorganic ion metabolism, including acidemia and hyperphosphatemia, although excluded from our definition of uremia, undoubtedly also contribute to untoward clinical manifestations of kidney failure. Given these difficulties, it is not surprising that although we have identified many uremic solutes, we know relatively little about their specific toxicity. In a few cases, uremic abnormalities have been reproduced by the transfer of uremic serum or plasma to normal animals or addition of these factors to the media of cultured cells (Box 51.2). However, the role of specific individual solute(s) in causing the abnormalities remains uncertain.

## SOLUTES CLEARED BY THE KIDNEY AND RETAINED IN UREMIA

The etymology of uremia derives from the Greek *ouron* (urine) and *haima* (blood), or "urine in the blood" from waste products typically excreted in the urine, most prominently urea. The long list of solutes retained in uremia has been assembled in two ways. Initially, biochemists would find a substance in the urine and then look for it in the blood of uremic patients. Several dozen uremic solutes were identified in this way as the biochemical pathways of intermediary metabolism were worked out. Beginning in about 1970, improved analytic techniques including gas chromatography, mass spectroscopy, and high-performance liquid chromatography were used to identify additional uremic solutes.[6,14] Recent technical advances including proteomic and metabolomic screening methods are lengthening the list of putative uremic solutes. However, the challenge in determining which solutes are toxic remains. In general, the compounds that are present in the highest concentrations, and therefore were identified first, have been studied most extensively. Experiments showing that uremic signs and symptoms can be replicated by raising solute levels in normal persons or animals to equal those observed in uremic patients are, for the most part, lacking. When attempted, such experiments have generally demonstrated that the levels required to produce

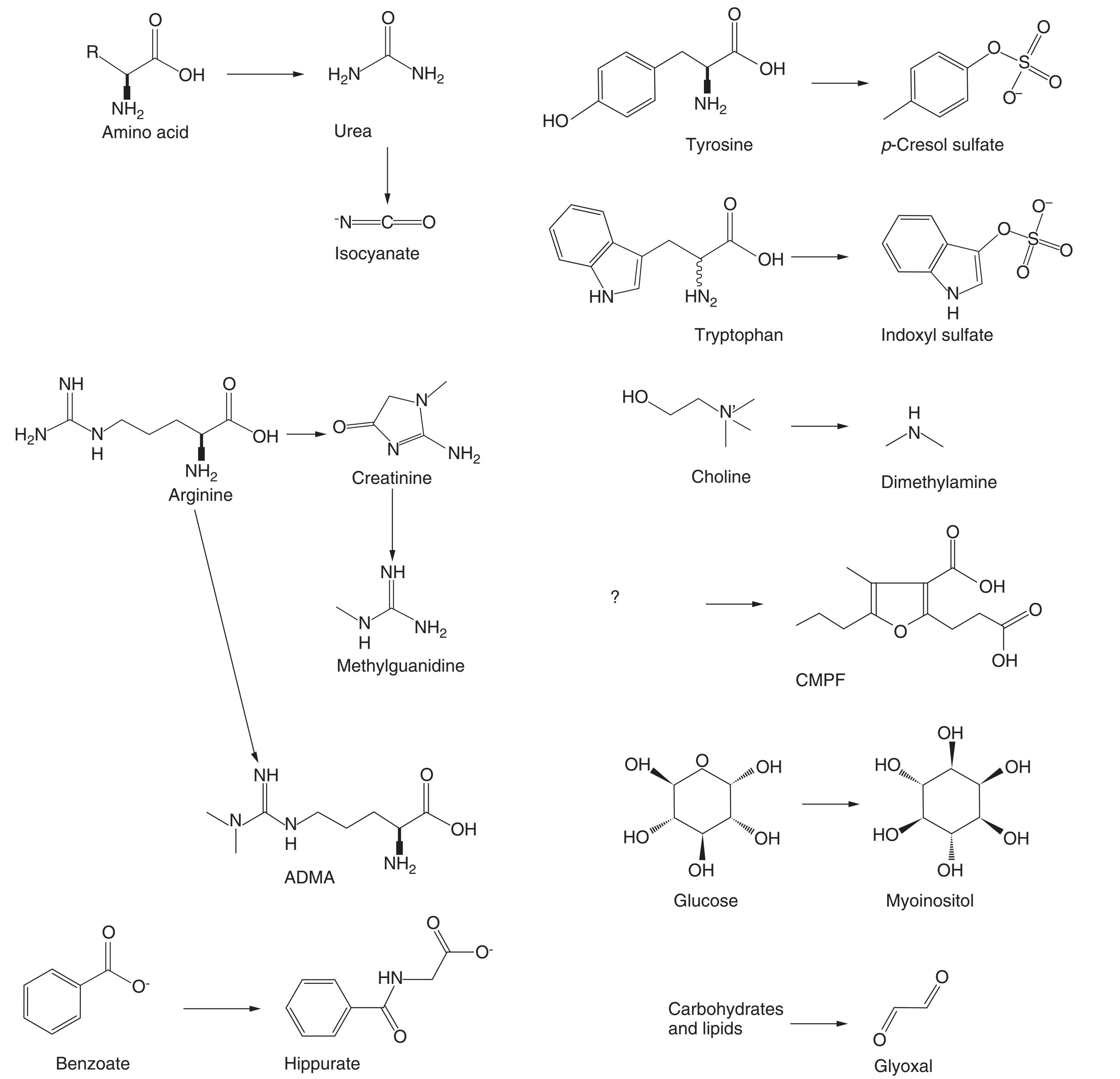


**Fig. 51.1 Generation of potential uremic toxins.** The substances in the *right column* of each panel are metabolites that are normally excreted by the kidney and therefore accumulate in the extracellular fluid when kidney function is lost. The *left column* shows the substances from which these potential "uremic toxins" are derived. In some cases, the biochemical derivation of the potential toxins is uncertain. For example, it is not known what fraction of the dimethylamine normally excreted is derived from choline, and the source of 3-carboxy-4-methyl-5-prophy-2-furanpr opanoic acid (CMPF) is obscure. See text for details. *ADMA,* Asymmetric dimethyl arginine.

**Box 51.2 Uremic Abnormalities Transferable With Uremic Serum or Plasma**

Inhibition of $Na^+,K^+$-ATPase
Inhibition of platelet function
Leukocyte dysfunction
Loss of erythrocyte membrane lipid asymmetry
Insulin resistance

*ATPase,* Adenosine triphosphatase.

toxic effects are higher than those measured in patients. Because so little is known about their toxicity, the discussion of uremic solutes is usually organized on the basis of their structure and not their contribution to symptoms or disease.

## INDIVIDUAL UREMIC SOLUTES

### UREA

Urea is quantitatively the most abundant solute excreted by the kidney, and its levels rise higher than those of any other solute when the kidney fails. The chemical formula for urea is $CH_4N_2O$, thus incorporating two nitrogen atoms per molecule of urea, so it is relevant to note that clinical laboratory assays traditionally report urea concentrations in terms of the nitrogen content of urea in the sample (blood urea nitrogen or BUN, most often reported in units of mg/dL or mmol/L). Early studies suggested that urea causes only a minor part of acute uremic illness.[15-17] In the most often cited of these studies, Johnson and colleagues[16] dialyzed patients with kidney failure against bath solutions containing urea. They found that initiation of hemodialysis with or without urea added to the dialysate (at 90 mg/dL) improved uremic symptoms including weakness, fetor, and gastrointestinal upset, even when urea was added. Furthermore, in patients already on dialysis, increasing the BUN level to 140 mg/dL (50 mmol/L) did not cause recurrence of uremic symptoms. Increasing the BUN level >140 mg/dL caused nausea and headaches, and increasing the BUN level >180 mg/dL (64 mmol/L) caused weakness and lethargy; however, symptoms in dialyzed patients whose BUN values were increased to these levels were less severe than symptoms in undialyzed patients with high BUN values. Studies in patients without kidney failure suggest that urea by itself does not cause uremia. Lastly, uremic symptoms have not been observed in patients in whom BUN levels are maintained at approximately 60 mg/dL (21 mmol/L) by high protein intake or increased tubular urea absorption.[18-20]

It is important to point out that these previous studies focused on the acute symptoms associated with elevated blood urea levels and not their chronic effects. The finding that uremia is not replicated by an isolated elevation of the plasma urea concentration does not mean that urea has no toxic effects.[21] Indeed, urea's dissociation product cyanate (Fig. 51.1) can trigger a nonenzymatic, posttranslational protein modification known as "carbamylation." The net result of the reaction is the addition of a "carbamoyl" moiety ($-CONH_2$) to a nucleophilic functional group (e.g., the primary amino group on the side chain of lysine), which can subsequently change the charge, structure, and function of a broad array of proteins (not surprising when considering the ubiquitous nature of urea in the body). [It must be noted that several authorities state that the reaction in question is more correctly deemed "carbamoylation," and, in fact, "carbamylation" denotes a different chemical reaction. Yet the biomedical literature has consistently described the addition of the carbamoyl moiety as "carbamylation," so we will follow this convention.]

Examples of pathophysiologic pathways accelerated by carbamylation include atherosclerosis, vascular calcification, and fibrosis of kidney parenchyma. Carbamylation of low-density lipoprotein (LDL) can increase its atherogenic properties by decreasing its binding to the LDL receptor[22] (preventing its clearance from circulation), a phenomenon seen in animal models[23,24] and in human CKD studies.[25-27] Carbamylated LDL has also been shown to trigger inflammatory signaling,[22,28,29] induce endothelial cell apoptosis,[30,31] and stimulate vascular smooth muscle proliferation.[32] Other studies demonstrate that carbamylated sortilin, elastin, uromodulin, and mitochondrial proteins all result in increased vascular pathology.[33-36] In animal models, carbamylated proteins at physiologic concentrations can activate glomerular mesangial cells to a profibrogenic phenotype[37] and result in renal tubular cell damage via stimulation of TGF-β, EGF, NF-κB, and endothelin-1.[38] Together, these data highlight how a single uremic retention solute (urea) can mechanistically result in multiple disease processes with significant clinical impact, with particular implications for vascular and myocardial disease. Indeed, numerous epidemiologic studies from different groups, employing different cohorts and even using different biomarkers of global carbamylation burden, have shown that carbamylation load is an independent risk factor for mortality, cardiovascular disease, heart failure, sudden cardiac death, and CKD progression (Table 51.1).

Carbamylated compounds, principally formed from protein exposure to urea, are considered among the top five uremic toxins with the highest toxicity evidence score.[10] Of note, to measure carbamylation, most studies measure a specific carbamylated protein (e.g., carbamylated albumin), protein-bound homocitrulline, or circulating free homocitrulline. L-homocitrulline is the amino acid that is produced when lysine is carbamylated on the terminal amino group of its side chain. Hydrolysis or proteolytic degradation of protein will release protein-bound homocitrulline, and measurement of the protein homocitrulline content or, alternatively, measurement of the homocitrulline-to-lysine ratio is a quantitative indicator of the relative amount of protein carbamylation of biological samples. Likewise, measurement of free homocitrulline in circulation also represents an indicator of total body carbamylation, as it presumably reflects the amount of protein that is carbamylated by urea and degraded during protein

**Table 51.1 Clinical Studies Linking Urea and Protein Carbamylation to Clinical Outcomes**

| Study Design | Sample Population | Outcomes | Ref. |
|---|---|---|---|
| Case control | 550 with preserved eGFR | MACE | 22 |
| Cohort | 187 on HD | Mortality | 41 |
| Case control | 366 on HD | Mortality | 42 |
| Cohort | 3111 with CKD | CKD progression, mortality | 43 |
| Cohort | 8040 | CKD progression | 189 |
| Cohort | 347 on HD | Mortality | 380 |
| Cohort | 111 with CHF | Mortality Cardiac transplantation | 381 |
| Cohort | 158 on HD | Erythropoietin resistance; mortality | 382 |
| RCT post-hoc analysis | 1161 on HD | CV death | 383 |
| Case control | 300 with CKD | CKD progression, mortality | 384 |
| Cohort | 1320 with CKD | CKD progression | 385 |

*CHF,* Congestive heart failure; *HD,* hemodialysis; *MACE,* major proteolytic turnover. Free and protein-bound homocitrulline thus both represent measures of protein carbamylation.[39,40] If urea is the driver of harmful carbamylation, one could speculate that interventions reducing urea such as dietary modification or intensification of hemodialysis treatments should yield benefits. However, clinical studies that indeed stood to modify urea levels have typically remained null. A likely reason is the complexity in targeting the putative pathway of harm. Individual measurements of blood urea levels, which represent protein intake, renal excretion, and overall nitrogen balance of patients at a single point in time, in clinical practice and research studies, do not appear to accurately reflect carbamylation toxicity. In contrast, carbamylation of proteins represents a time-averaged measure of urea levels over the entire lifespan of that protein. Possibly because of these differences in kinetics, studies in hemodialysis patients have observed that the mortality risks associated with carbamylation were significantly greater than, and independent of, the risks associated with blood urea levels.[41] Furthermore, a case (mortality within 1 year of starting dialysis) versus control (demographically matched survivors of >1 year on dialysis) study showed that while urea, dialysis clearance of urea measured by Kt/V, and protein catabolic rate were similar, the distribution of carbamylated albumin levels was significantly higher in subjects who died compared with survivors (Fig. 51.2)[42] In nondialyzed CKD patients, carbamylated protein levels strongly correlate with blood urea concentrations and both carbamylated albumin and blood urea levels are associated with increased mortality; despite their correlation, however, the mortality risk associated with carbamylated albumin is still significant even after adjusting for patients' blood urea levels, suggesting that carbamylation is a function of more than just urea levels.[43] In fact, it has been shown that carbamylated albumin levels in hemodialysis patients also correlated negatively with circulating concentrations of free amino acids, suggesting that protein-energy wasting and amino acid deficiencies associated with CKD may also contribute to protein carbamylation and its associated survival risk.[41,44] Lastly, the association between carbamylation and the sequelae of uremia is further suggested by the fact that when end-stage kidney failure (ESKF) patients were transitioned from standard intermittent hemodialysis to extended duration nocturnal hemodialysis (double the treatment time), it caused dramatic reduction in protein carbamylation in many patients and the reduction in carbamylation correlated with reductions in left ventricular hypertrophy and other favorable outcomes.[44]

Such discrepancy between the downstream toxic effect of urea (carbamylation) and simple blood urea measurements may explain why prior studies in dialysis and CKD employing interventions that lowered urea did not show the effects that would be expected had carbamylation specifically been targeted.[45] Although the clinical risks associated with protein carbamylation combined with the basic experiments showing pathologic effects of carbamylation on cellular

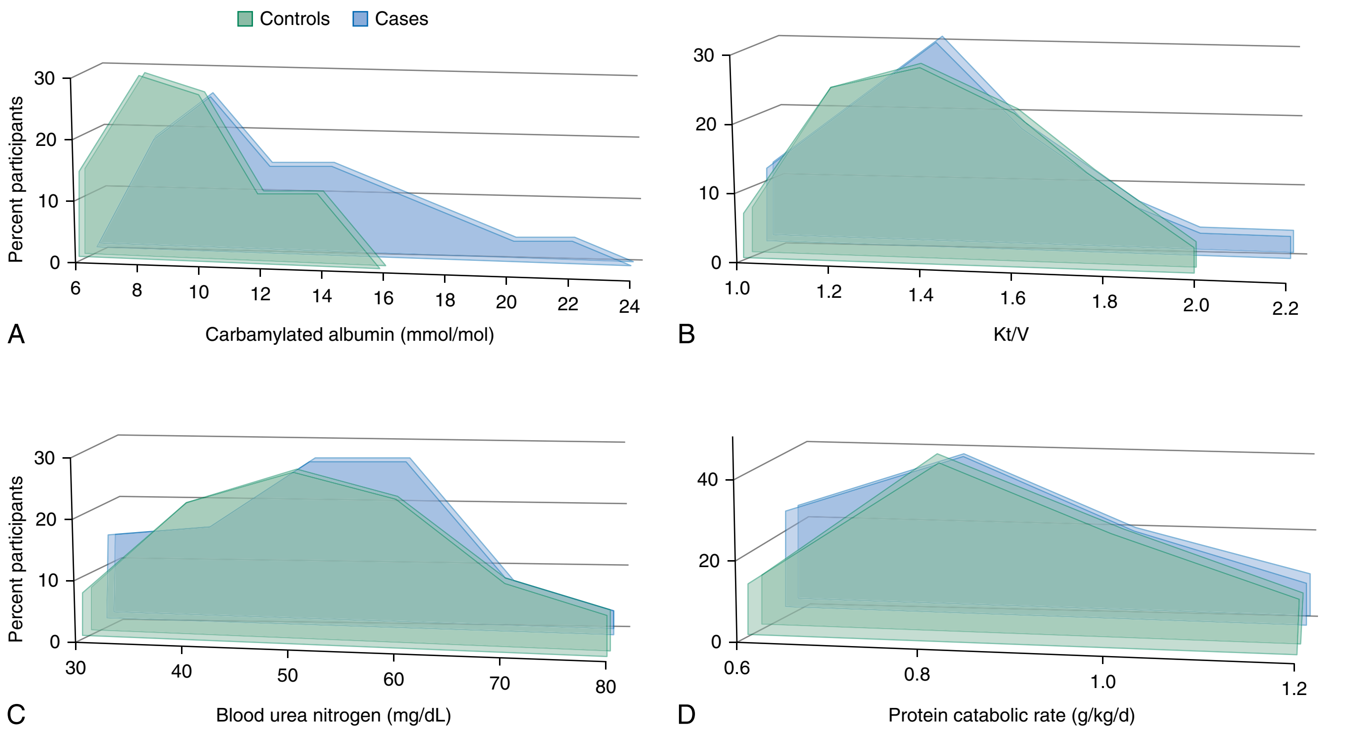


**Fig. 51.2 Patient and control distributions of carbamylated albumin, Kt/V, blood urea nitrogen (BUN), and normalized protein catabolic rate.** (A) Distribution of carbamylated albumin level measures in the study population by patients (blue; $n = 122$ who died within 1 year of starting dialysis) and matched controls (green; $n = 244$ who survived the first year of dialysis); $P = 0.01$). (B) Distribution of the average Kt/V measure in the study population by patients (blue) and controls (green; $P = 0.55$). (C) Distribution of the average BUN measure in the study population by patients (blue) and controls (green; $P = 0.51$). (D) Distribution of normalized protein catabolic rate measured in the study population by patients (blue) and controls (green; $P = 0.15$). $P$ values represent the frequency distribution comparisons. (Reproduced from Kalim S, Trottier CA, Wenger JB, et al. Longitudinal changes in protein carbamylation and mortality risk after initiation of hemodialysis. *Clin J Am Soc Nephrol.* 2016;11:1809–1816.)

functions provide intriguing evidence, further investigation is needed to prove its causal role in the pathophysiology of uremia.

### D–AMINO ACIDS

In comparison with urea, we know much less about most other potential uremic toxins. The D–amino acids exemplify this problem. Aggregate plasma concentrations of D–amino acids increase as kidney function declines.[46,47] However, the source, clearance, and toxicity of the D–amino acids found in the plasma are not well defined. D–Amino acids can be synthesized by mammalian cells, as well as derived from food, or produced by colonic bacteria.[48] Circulating D–amino acids are filtered by the glomerulus and then, in varying proportion, reabsorbed intact, degraded by D–amino acid oxidase (DAO) or D–aspartic acid oxidase in the proximal straight tubule, or excreted unaltered in the urine.[49,50] The liver can also clear D–amino acids, but the relative importance of renal and hepatic clearance is unknown. Aggregate D–amino acid concentrations have been found to increase almost in proportion to the serum creatinine level in kidney failure, suggesting that renal clearance predominates.[46,51,52] However, measured concentrations of individual D–amino acids, such as D-serine, increase less than creatinine.[51,52] This discrepancy remains unexplained. Of interest, D-serine is an endogenous coagonist of the synaptic *N*-methyl-D-aspartate receptor. It is tempting to speculate that D–amino acids are cleared rapidly from the extracellular fluid (ECF) because they have toxic effects. In addition, it has long been presumed that high levels of D–amino acids could impair protein synthesis or function.[48] D–amino acid accumulation could also interfere with the effects of endogenous D-serine and D-alanine on neuronal function,[53] but no major ill effects of D–amino acid accumulation have been observed in DAO-deficient mice, which have higher D–amino acid levels than humans with impaired kidney function.[54,55] Exogenous D–amino acids have so far been shown to be toxic only when administered in large quantities.[54,56]

### PEPTIDES AND PROTEINS

The kidney clears circulating dipeptides and tripeptides, which may comprise a significant portion of the extracellular amino acid pool.[57] Filtered dipeptides and tripeptides can be broken down by brush-border peptidases and reabsorbed as amino acids or reabsorbed by a brush-border peptide transporter and then hydrolyzed within proximal tubule cells.[58] Peritubular uptake, again followed by hydrolysis to amino acids, makes the renal clearance of many peptides higher than the glomerular filtration rate (GFR).[57,59] Small peptides are also taken up by other organs and generally do not accumulate in kidney failure. Peptides containing altered amino acids, which are normally cleared by the kidney, may be an exception to this rule.[59]

The kidney plays a proportionally larger role in the clearance of larger peptides. Proteins with a molecular weight of 10 to 20 kDa, such as $\beta_2$-microglobulin and cystatin C, are normally filtered by the glomerulus and then endocytosed and hydrolyzed in the lysosomes of proximal tubular cells.[60,61] Their plasma concentrations therefore rise in parallel with the plasma creatinine level as kidney function declines. Indeed, the plasma concentration of cystatin C, which is released at a near-constant rate by nucleated cells, may yield a more reliable estimate of GFR than the concentration of creatinine.[62] The role of the kidney in the removal of peptides with molecular weight between 500 Da and 10 kDa is less well defined. Peptides in this range are also filtered by the glomerulus and then hydrolyzed by brush-border peptidases or endocytosed, depending on their size and structure. Biologically active peptides such as insulin may also be cleared by peritubular uptake. Studies in patients with inherited dysfunction of proximal tubular endocytosis suggest that the normal kidney clears approximately 350 mg/day of peptides with molecular weights of 5 to 10 kDa from the circulation.[63] The relative importance of renal to extrarenal clearance has not been defined for most substances in this size range. The extent to which circulating levels of such peptides are increased in kidney failure is therefore unpredictable. Even less is known about the kidneys' contribution to the clearance of peptides in the range of 500 Da to 5 kDa.

Although the aggregate peptide levels in kidney failure remain ill defined, we have some knowledge of individual peptides that are retained in uremia. The best known are small proteins or fragments of large proteins for which immunoassays have been developed.[64-78] Box 51.3 provides a partial list of these substances. Retinol-binding protein, $\alpha_1$-microglobulin, and β-trace protein (also known as prostaglandin $D_2$ synthase) are members of the lipocalin superfamily, and future studies may identify elevated levels of other proteins of this group. Lipocalins are a family of proteins that transport small hydrophobic molecules and have an eight-stranded, antiparallel, symmetrical β-barrel fold. This reflects a β sheet that has been rolled into a cylindrical shape. A ligand-binding site is present in this cylinder. Studies using proteomic techniques have yielded a more complete picture of individual peptides that are retained in uremia.[79-81] These studies have shown that, as expected, uremic plasma contains a vast array of protein fragments that are normally cleared by the kidney. Many of these are derived from fibrinogen and the complement cascade.[81,82] One study has identified >1000 peptides with molecular weights from 800 Da to 10 kDa in the plasma of patients undergoing dialysis.[83] The toxicity of these

**Box 51.3 Low-Molecular-Weight Proteins and Protein Fragments That Accumulate in Uremia**

$\alpha_1$-Microglobulin
$\beta_2$-Microglobulin
β-Trace protein (also known as prostaglandin D2 synthase)
Clara cell protein
Chromogranin A
Cystatin C
Free immunoglobulin light chains
Natriuretic peptides
Procalcitonin
Retinol-binding protein
Transcobalamin
Leptin
Ghrelin
Resistin
Troponin I

peptides and low-molecular-weight proteins, like that of smaller molecular uremic solutes, is largely unknown. It has been widely speculated that retained peptides can cause inappropriate activation of various hormone or cytokine receptors. For example, retained complement protein D (molecular weight, 24 kDa) could contribute to systemic inflammation and accelerated vascular disease observed in patients receiving dialysis.[84] Complement factor D is a serine protease crucial for activating the alternative pathway by cleaving complement factor B (FB) to generate the C3 convertases C3($H_2O$)Bb and C3bBb. Complement factor D is primarily produced by adipose tissue and circulates in its active form. The overall hypothesis that retained peptides contribute to uremia remains to be proven except for the case of $\beta_2$-Microglobulin. $\beta_2$-Microglobulin amyloidosis is the only disease attributable with certainty to a retained peptide, and its prevalence among the dialysis population appears to have declined over the past decades.[85] Plasma levels of peptides such as procalcitonin, troponin I, N-terminal brain natriuretic peptide, and chromogranin A are, however, increasingly used as diagnostic markers. And "false-positive" results may be obtained if criteria are not adjusted for the elevation of their plasma levels, which normally accompanies loss of kidney function.

## GUANIDINES

Among the compounds most frequently considered uremic toxins are guanidines, which, like urea, are derived from arginine (see Fig. 51.1).[86-88] One group of guanidines that accumulate in uremia includes creatinine and its breakdown products. Creatinine is produced by nonenzymatic conversion of creatine, which in turn is made from guanidinoacetic acid.[89] Creatinine itself appears nontoxic, and levels have been increased transiently to >100 mg/dL (8800 μmol/L) in subjects undergoing clearance studies. Instead, interest has been focused on the potential toxicity of various creatinine metabolites, especially creatol and methylguanidine (MG).[90,91] Creatol is a product resulting from the reaction of creatinine with the hydroxyl radical and is identified as a precursor of MG. Creatol and MG levels normally approximate 1% of creatinine levels. The production of these substances increases as plasma creatinine concentrations rise and may be stimulated by increased levels of intracellular oxidants.[87,89,90] MG is also produced by colonic bacteria, and its production may be increased by augmented dietary intake of protein or creatinine.[92] Guanidinosuccinic acid (GSA) is a nitrogenous metabolite that is found in elevated levels in both serum and urine of patients with advanced CKD. GSA is a guanidine formed not from creatinine but from the urea cycle intermediate argininosuccinate.[93,94] Rising plasma urea concentrations impede the conversion of argininosuccinate to urea and increase the production of GSA. The production of GSA thus depends on dietary protein intake, as well as kidney function, and may also be stimulated by increased concentrations of intracellular oxidants.[94,95]

Creatol, MG, and GSA share the interesting property that their plasma concentrations rise out of proportion to urea and creatinine levels as the GFR declines. This is because they are cleared largely by the kidney, and their production rates increase when plasma creatinine and urea concentrations are elevated.[87,89,90] In addition, relative to creatinine, large volumes of distribution, combined with restricted intercompartmental diffusion, limit the removal of creatol, MG, and GSA by intermittent hemodialysis.[86] In patients undergoing conventional hemodialysis, these compounds therefore exhibit high concentrations relative to normal.[4] The finding that they are present in relatively high concentrations does not prove that they are toxic. However, the evidence for toxicity of various guanidines, although incomplete, is stronger than that for most other solutes. Administration of MG aggravates uremic symptoms in dogs, whereas GSA contributes to uremic platelet dysfunction, and a number of guanidines impair neutrophil function.[96-98] In addition, various guanidines have been shown to accumulate in the brain and cerebrospinal fluid in uremia and may contribute to central nervous system (CNS) dysfunction.[99]

The methylated arginines, asymmetric dimethyl arginine (ADMA) and symmetric dimethyl arginine (SDMA), also accumulate in kidney failure (see Fig. 51.1). The metabolism of methylated arginines is quite different from that of other uremic guanidines. ADMA and SDMA are formed by the methylation of arginine residues in nuclear proteins, which are released when these proteins are degraded. Interest has focused largely on ADMA because it inhibits nitric oxide (a potent endothelial-derived vasodilator) synthesis by the endothelial nitric oxide synthase (eNOS). The enzyme eNOS converts L-arginine to nitric oxide, and L-citrulline and is an endothelial-enriched messenger RNA and protein. ADMA inhibits eNOS enzymatic activity.[100,101] SDMA has been less extensively considered as a toxin, but recent evidence has associated its plasma levels with cardiovascular risk.[101] The urinary clearance of ADMA is similar to that of creatinine, but most plasma ADMA is taken up and degraded in various tissues including the kidney.[100,102] The presence of nonrenal clearance limits the rise in plasma ADMA concentrations observed as kidney function declines, so ADMA concentrations rise to approximately two times normal early in the course of CKD and then do not increase much further as patients advance to ESKF.[102,103] Increases in plasma ADMA concentration, although modest in proportion to other uremic solutes, have been associated with an increased risk for cardiovascular disease, especially through impaired bioactive NO function, and death in patients with CKD.[101] It should be noted that differences in assay methods and reported reference ranges for ADMA greatly complicate the interpretation of these studies.

## PHENOLS AND OTHER AROMATIC COMPOUNDS

Phenols are compounds that have one or more hydroxyl groups attached to a benzene ring. In discussions of uremia, phenols are usually considered together with other aromatic compounds, such as hippurates, and the term "phenols" is sometimes used loosely to include these other substances. The aromatic compounds normally found in the ECF compartment are, for the most part, derived from the amino acids tyrosine and phenylalanine or from aromatic compounds contained in dietary vegetables. Medications provide an additional source for patients. The compounds in the ECF are mostly metabolites; these are derived from their parent compounds by a combination of methylation, dehydroxylation, oxidation, reduction, and/or conjugation. Many of these reactions take place in colonic bacteria.

The final step, which is usually conjugation with sulfate, glucuronic acid, or an amino acid, may take place in the liver, intestinal wall, or, to a lesser extent, kidney.[104,105] In general, conjugation tends to make the aromatic compounds both less toxic and more polar, thus facilitating their excretion by various organic ion transport systems.

These metabolic processes produce a bewildering array of aromatic compounds that are normally excreted in the urine or feces. The aggregate urinary excretion of aromatics is about 1000 mg/day and varies widely with the diet. The compounds normally excreted by the kidney accumulate in uremia and contribute to elevation of the anion gap, because most aromatic conjugates are negatively charged.[106] The concentration of individual aromatic compounds in uremic patients ranges from barely detectable up to 500 μM.[4,107-109] The relatively few compounds that have been studied extensively, including the examples described as follows, are among those found in the highest concentration. Interest in the contribution of phenols and other aromatic compounds to uremic toxicity has been encouraged by reports that uremic symptoms are better correlated with plasma concentration of these compounds than with those of other solutes.[17,110-112] Evidence obtained so far on the toxicity of individual aromatic compounds is incomplete.

The most extensively studied aromatic uremic solute is hippurate (see Fig. 51.1). Hippuric acid (Gr. hippos, horse, ouron, and urine) is the aromatic waste compound normally excreted in the largest quantity. The concentration of free hippurate rises higher than those of other aromatic solutes in the plasma of patients with kidney failure. Hippurate is the glycine conjugate of benzoate, derived largely from vegetable foods, with only a small amount formed endogenously from the amino acid phenylalanine.[113,114] Levels of hippuric acid rise with the consumption of phenolic compounds (such as in fruit juice, tea, and wine). Diet therefore determines hippurate production, and hippurate excretion in aboriginal people eating vegetable diets may exceed hippurate excretion in people from industrialized nations by many fold.[115] In persons with normal kidneys, active tubular secretion maintains a plasma hippurate concentration much lower than it would be if hippurate were cleared solely by glomerular filtration. There is a clinical correlate of this property of hippurate. Aminohippuric acid or *para*-aminohippuric acid (PAH), a derivative of hippuric acid, is a diagnostic agent used in the measurement of renal plasma flow. PAH is completely removed from the blood that passes through the kidneys given that PAH undergoes both glomerular filtration and tubular secretion. Therefore the rate at which the kidneys can clear PAH from the blood reflects total renal plasma flow. By itself, hippurate is not toxic. The plasma hippurate concentration in a normal person can be increased to equal that of a uremic patient, without apparent ill effect.[116] Moreover, increasing the hippurate concentration by benzoate feeding in a patient with kidney failure does not aggravate uremic symptoms.[117] A further clinical correlate addresses the potential toxicity of aromatics, in this case, an exogenous source. Toluene is a widely abused inhaled drug (e.g., glue sniffing) due, in part, to its acute neurologic effects of euphoria and hallucinations. Toluene is metabolized by cytochrome P-450 into benzoic acid and hippuric acid. The hallmarks of acute toluene intoxication are hypokalemic paralysis and metabolic acidosis due to distal renal tubular acidosis. Patients can also have hippurate crystals appear in the urine. They can appear either yellow-brown or clear and frequently take the form of needle-like prisms or plates.

Another extensively studied aromatic compound is *p*-cresol. In contrast to hippurate, which is derived from aromatic compounds in plants, *p*-cresol is formed by the action of colonic bacteria on tyrosine and phenylalanine. The portion of amino acids that escapes absorption in the small intestine may be increased in uremic patients, leading to increased production of *p*-cresol and other bacterial metabolites.[118,119] Studies have shown that *p*-cresol circulates largely as *p*-cresol sulfate with a much lesser portion in the form of *p*-cresol glucuronide. Reports of unconjugated *p*-cresol in the plasma of uremic patients now appear to have been the result of inadvertent hydrolysis of *p*-cresol sulfate during sample processing.[120,121] *p*-Cresol sulfate binds avidly to serum albumin, and the effect of different kidney replacement therapies on albumin-bound solutes has often been tested by measuring plasma *p*-cresol sulfate concentrations.[122-124] High concentrations of *p*-cresol sulfate (often measured as *p*-cresol) have been associated with cardiovascular death in patients undergoing hemodialysis, and *p*-cresol sulfate has been related to indices of endothelial injury, including endothelial microparticle production.[125-127] While the results of clinical studies are not uniform, these findings, along with in vitro evidence of *p*-cresol sulfate toxicity, have increased the focus on *p*-cresol sulfate as a potential uremic toxin.[126-128]

Other aromatic uremic solutes have been identified in great numbers but studied less extensively.[11,108,109] Metabolites of tyrosine and phenylalanine, which also accumulate in uremia, include phenlylacetylglutamine, *para*-hydroxyphenylacetic acid, and 3,4-dihydroxybenzoic acid.[129-131] Phenylacetylglutamine, like *p*-cresol sulfate, has been associated with cardiovascular disease and mortality in dialysis patients.[132] The structural relationship of the aromatic amino acid metabolites to neurotransmitters has stimulated interest in their potential role as uremic neurotoxins. So far, 3,4-dihydroxybenzoate has been shown to cause CNS dysfunction in rats, but only at levels higher than those encountered in patients with kidney failure.[130] Increased levels of 4-hydroxyphenylacetate were associated with impaired cognitive function in patients maintained on hemodialysis.[133] The work of identifying toxic aromatic uremic solutes, however, remains daunting. Relatively little progress has been made, and caution is appropriate, particularly when interpreting studies relating levels of specific compounds to poor outcomes. Associations have necessarily been identified only for the few compounds extensively studied. Even if a positive association has been correctly identified, correlations of poor outcomes with abnormal blood levels could be confounded by associations with other unmeasured aromatic substance(s) evidencing related production and/or clearance rates.

## INDOLES AND OTHER TRYPTOPHAN METABOLITES

Indoles are compounds containing a benzene ring fused to a five-membered nitrogen-containing pyrrole ring (see Fig. 51.1). Compounds with an indole ring structure are hydrophobic and, hence, are highly protein bound in the circulation. This makes them difficult to filter at the kidney glomerulus or remove with therapeutic dialysis via diffusion. Thus some indoles are waste products that accumulate when kidney function is impaired and have been considered to be, in part, uremic toxins. Many similarities are encountered when considering the indoles and phenols in uremia. As with phenols, some indoles are derived from plant foods and others are produced endogenously. However, the endogenous indoles are derived mostly from tryptophan, whereas the phenols are derived from phenylalanine and tyrosine. Diverse chemical modifications of L-tryptophan yield a remarkable variety of tryptophan-derived indole structures; more than 600 indoles are known. As with the phenols, minor chemical modifications in various combinations yield a remarkable variety of structures.[134] Those with known physiologic function include the neurotransmitter 5-hydroxytryptamine (serotonin) and melatonin. Other indoles are considered to be waste products and are often conjugated before urinary excretion. These uremic indoles accumulate when kidney function is impaired.

The most extensively studied of the uremic indoles is indoxyl sulfate; this is produced from tryptophan in a manner reminiscent of the production of *p*-cresol sulfate from tyrosine and phenylalanine. Gut bacteria convert tryptophan to indole, which is then oxidized to indoxyl and conjugated with sulfate in the liver. There is evidence that indoxyl sulfate is toxic in vitro and in animal studies, but early studies of indoxyl sulfate infusion failed to replicate uremic symptoms.[17,135-138] Like *p*-cresol sulfate, indoxyl sulfate is extensively bound to plasma albumin. It has also been suggested that indoxyl sulfate is toxic to renal tubular cells, and the rising indoxyl sulfate levels accelerate the progression of renal disease.[139] Controlled trials to lower indoxyl sulfate levels, however, have not been found to slow progression.[140]

Other indoles that accumulate in uremia include indoleacetic acid, indoleacrylic acid, and 5-hydroxyindoleacetic acid.[4,141,142] As with the phenols, indoles are structurally related to potent neuroactive substances, including serotonin and (famously) D-lysergic acid diethylamide. This structural similarity has stimulated interest in the potential role of indoles as neurotoxins, but few uremic indoles have been administered to normal animals and none have convincingly been shown to alter CNS function at the levels encountered in patients with kidney failure. There is, however, increasing evidence that indoxyl sulfate contributes to vascular disease and particularly to thrombosis.[127,143] Some effects of indoxyl sulfate have been shown to be mediated by activation of the aryl hydrocarbon receptor, a transcription factor originally identified as upregulating xenobiotic disposal pathways.[143,144] This work puts our knowledge of the mechanisms by which indoxyl sulfate acts ahead of that of other putative uremic toxins.

Only a minor portion of dietary tryptophan is excreted as indoles. Most is metabolized by the kynurenine pathway, which allows tryptophan to be converted to glutarate and oxidized or, when necessary, used in the synthesis of nicotinamide. Kidney failure causes members of the kynurenine pathway, including L-kynurenine and quinolinic acid, to accumulate in the plasma.[145,146] Knowledge that these substances play a physiologic role in the modulation of CNS function has stimulated interest in their possible contribution to uremic toxicity. As usual, however, evidence that they are toxic at the levels encountered in patients has not been obtained.

## ALIPHATIC AMINES

The methylamines monomethylamine (MMA), dimethylamine (DMA), trimethylamine (TMA), and trimethylamine oxide (TMAO) are among the simplest compounds that have been considered to be uremic toxins. Reported serum levels are twofold to threefold higher in patients with ESKF compared with persons with normal or near-normal kidney function.147,148 However, available data and predictions based on their chemistry suggest that the methylamines are poorly removed by dialysis, and early data suggest that they may even be produced in excess in those with uremia.147-149

A large volume of distribution may contribute to poor removal of the methylamines by dialysis. These compounds are bases, with a $pK_a$ ranging from 9 to 11. Thus they exist as positively charged species at a physiologic pH. The lower intracellular pH compared with extracellular pH should lead to their preferential intracellular sequestration, with volumes of distribution exceeding total body water. Indeed, measurements in experimental animals and humans have confirmed these predictions for both DMA and TMA.[148,150,151]

Because they circulate as small organic compounds that are not protein bound, these three amines are likely freely filtered. However, because they exist as organic cations, they also have the potential to be secreted by one or another of the family of organic cation transporters and may also travel through Rh channels.[152,153] Hence they may achieve clearances that are higher than the GFR. The chemically similar exogenous compound, tetraethyl ammonium, has long been a prototype test solute for organic cation secretion and is cleared at rates up to (and in one study higher than) the renal plasma flow.[147,154] Although formal renal clearances of DMA and TMA are not available, the total metabolic clearance of DMA and TMA by plasma disappearance of labeled compounds in rats approaches that of renal plasma flow.[150] By contrast, the urinary clearance of MMA in normal subjects is about one-third that of creatinine, indicating no net secretion for this amine.[149]

The biochemical pathways leading to MMA, DMA, and TMA are not well delineated. Both the host's mammalian tissues and resident gut flora have been thought to contribute to the net appearance of these amines. However, plasma MMA and DMA concentrations were not different among patients with ESKF with and without colons.[155] The dietary precursors for MMA, DMA, and TMA include choline, carnitine, betaine, and dimethylglycine.[156-158] Production of these compounds may actually be increased with kidney failure, perhaps caused by overgrowth of intestinal bacteria.[149,151,159] Thus production of aliphatic amines may be increased in the face of impaired renal removal.

Incomplete data also implicate the amines as toxic. Reported effects of MMA include a variety of neural toxicities, hemolysis, and inhibition of lysosomal function.[160] MMA has also been shown to be a potent anorectic agent when administered into the cerebrospinal fluid in mice.[161] Despite toxicities in cells and animals, however, MMA and DMA were not associated with all-cause mortality in a cohort of patients with ESKF.[162] Although of utmost importance, mortality is obviously a blunt metric of uremic toxicity. Other signs of toxicity of MMA and DMA should be explored.

The uremic fetor or fishy breath noted in uremic patients is attributable to TMA.[163] Although the malodor may be of no major consequence in itself, the potentially important and well-described diminutions in taste (dysgeusia) and smell (dysosmia) among patients with kidney failure (which may contribute to poor nutritional status) may also be related to the amines. Plasma MMA concentrations were not related to olfactory defects in ESKF.[164]

TMA can be both a precursor to TMAO and, as noted, a product of dietary TMAO.[165] TMAO is cleared by secretion in the normal kidney; its levels rise as the GFR falls and, like those of many other secreted solutes, are increased out of proportion to urea levels in dialysis patients.[166] TMAO was initially identified as a risk factor for cardiovascular disease in persons with normal renal function.[165,167] TMAO is a gut microbial metabolite that has been shown to be directly atherogenic. TMAO arises from gut microbiota metabolism following ingestion of diets rich in phosphatidylcholine (lecithin), the major dietary source of choline, and carnitine, an abundant nutrient in red meat. Studies employing microbial transplantation into recipients have confirmed a direct causal role for gut microbes in transmitting atherosclerosis susceptibility and overall TMAO production. Subsequent studies showed that accumulation of TMAO is also associated with atherosclerosis burden and cardiovascular events in patients with renal insufficiency and in patients maintained on dialysis.[168,169]

### OTHER UREMIC SOLUTES

A wide variety of other compounds accumulate in kidney failure. One group is the polyols, of which the most extensively studied is myoinositol (see Fig. 51.1).[170,171] Myoinositol is different from most other uremic solutes in that it is normally oxidized by the kidney. Its accumulation in uremia therefore reflects impaired degradation and not impaired excretion. Evidence that myoinositol causes nerve damage, although stronger than most of the evidence for the toxicity of uremic solutes, is far from conclusive.[172]

The purine metabolite uric acid is the only known organic substance with a plasma level actively regulated by variation of its renal excretion. With advanced CKD, the capacity of the kidney to increase the fractional excretion of uric acid is exceeded and uric acid levels increase, along with those of its precursor molecules, xanthine, and hypoxanthine. Observational studies have shown that high uric acid levels are associated with increased survival in dialysis patients, perhaps because the plasma uric acid is a marker of nutritional status.[173] Other nucleic acid metabolites excreted by the kidney are produced in much lesser quantities. Many are derived from the modified nucleosides contained in transfer RNAs.[174] They appear to be cleared largely by filtration and to accumulate in the plasma as the GFR falls. Pseudouridine is an isomer of the nucleoside uridine in which the uracil is attached via carbon–carbon instead of a nitrogen–carbon glycosidic bond. It has been suggested that pseudouridine, which is the most abundant of these variant nucleic acid metabolites, contributes to insulin resistance and altered CNS development but, as usual, the demonstration of its toxicity is not conclusive.[174,175]

Oxalate is also excreted by the kidney and accumulates in kidney failure. Oxalate is derived from catabolism of endogenous substances including vitamin C, as well as from plant foods.[176,177] The potential for oxalate deposition in tissues may limit our ability to maintain normal plasma vitamin C concentrations in patients receiving dialysis.[178,179] Additional substances excreted by the kidney that accumulate in kidney failure include various pteridines, dicarboxylic acids, isoflavins, and furancarboxylic acids including 3-carboxy-4-methyl-5-prophy-2-furanpropanoic acid.[123,174,180-183]

Indeed, studies continue to add new solutes to the list and the number reported increases rapidly with upgrades in metabolite screening methods. This list includes known and now even more unknown compounds that are at present identified only by their molecular mass using mass spectrometry and do not yet appear in standard databases of human metabolites. The possibility of toxicity is invariably considered when new solutes are identified, but experiments to test the solute toxicity are rarely performed. These challenges are common to metabolomic and proteomic studies in general but may be exacerbated in nephrology research.[184] Within the realm of metabolomic kidney disease research, a well-recognized challenge is that a substantial portion of all detectable metabolites track with estimated glomerular filtration rate (eGFR), creating problems of confounding when making associations involving eGFR. While adjusting for eGFR's influence on metabolites is considered necessary, analytes that strongly correlate with eGFR may still be mechanistically important and worthy of additional study.[185] The sheer number of metabolites that can be assessed using modern technologies increases the risk of false associations due to chance alone from multiple testing. New biostatistical approaches and analytical techniques aim to reduce these risks, but replication remains an important burden for putative uremic toxins.[186,187] External validation of findings through replication studies with large and sufficiently powered sample sizes thus remains a critical component of metabolomic research, but meeting such methodologic demands in the setting of specific populations or phenotypes of interest where data can be limited is difficult.[188] Translating findings for candidate markers from high throughput screening studies is even more difficult.[189] These dilemmas highlight why progress in this field remains incremental.

## SOLUTE REMOVAL BY DIFFERENT FORMS OF KIDNEY REPLACEMENT THERAPY

Although investigators have not succeeded in replicating uremic illness by administering potential uremic solutes to normal humans or animals, reversing illness by removing solutes has become a part of everyday practice. Because renal replacement therapies remove solutes indiscriminately, the improvement they effect cannot be attributed to removal of specific compounds. However, the different forms of kidney replacement therapy clear solutes at different rates based on some defining characteristics including molecular size, protein binding, and sequestration within cells or other body compartments. The demonstration that different therapies have different effects on some features of uremia might therefore reveal properties of the responsible toxin(s). While randomized human studies are needed to definitively assign or refute a direct causal role for any given solute and a clinical uremic sign or symptom, the ability to modulate putative toxins is a prerequisite. To date, studies targeting urea kinetics, using higher-flux dialyzers, and increasing the frequency and duration of hemodialysis treatments have shown these strategies to be ineffective at removing several well-described retention solutes, particularly protein-bound ones (discussed later).[190-192] For further discussion on kidney replacement therapies, please see Chapters 62 and 63.

## ORIGINAL MIDDLE MOLECULE HYPOTHESIS

The suggestion that the nature of uremic toxins could be deduced by comparing the effects of different renal replacement methods was first advanced by Babb and colleagues.[193] In the 1960s, hemodialysis was performed with membranes that provided limited clearance of solutes with molecular weight >1000 Da. Treatment with these membranes wakened patients from coma, relieved vomiting, and partially reversed other uremic symptoms. This provided evidence, which remains convincing, that some important uremic toxins are small. Babb and coworkers were impressed that patients on peritoneal dialysis were healthier than patients on hemodialysis who had the same plasma urea and creatinine concentrations. They further observed that increasing the dialysis duration from 6.5 to 9 hours three times weekly prevented neuropathy. These observations led them to conclude that important toxins were >300 Da because, as compared with contemporary hemodialysis membranes, the peritoneal membrane afforded greater relative permeability in this size range and because increasing the hemodialysis session length was expected to reduce the plasma concentration of larger molecules more than the concentrations of creatinine and urea. On the basis of their further impression that no additional benefit was obtained using membranes that provided superior clearance for solutes >2000 Da, they concluded that some important toxins were "middle molecules," with a molecular weight >300 Da but <2000 Da.[194]

## LARGE SOLUTES—CHANGING DEFINITION OF "MIDDLE MOLECULES"

Studies during the 1970s provided only equivocal evidence that increasing the clearance of solutes with a molecular weight between 350 and 2000 Da improved the health of uremic patients.[193] The proposition that no benefit could be obtained by increasing the clearance of solutes with molecular weight >2000 Da was never prospectively tested. The original middle molecule hypothesis was thus never proven to be correct. And although the phrase "middle molecules" remains in use, its meaning has gradually shifted to include larger solutes. The 2003 report of the EUTox work group[4] thus defined middle molecules as those with a size ranging from 500 Da to <60,000 Da. The adoption of new membrane materials, which was in part a response to the original middle molecule hypothesis, ended the investigation of the relative toxicity of solutes that fall into different parts of the size range <1000 Da. One problem is that the concentration of putative high-molecular-weight toxins may not decline in proportion to the increase in smaller molecule clearance during treatment because they are cleared by extrarenal mechanisms and they move slowly from the interstitial fluid to the ECF during treatment.[196,197]

To improve the clearance of the so-called middle molecules, hemodiafiltration (HDF) has been established, adding convection to diffusion and generating higher hydrostatic pressure differences across the membrane. The majority of studies in this arena focused on the different dialysis modalities and their impact on clinical outcomes such as mortality, major cardiovascular events, hospitalization, and treatment-related adverse events. For example, a Cochrane review including 35 studies enrolling more than 4000 patients demonstrated that compared with conventional hemodialysis, HDF reduced cardiovascular events but did not have an impact on all-cause mortality.[198] Then, in 2023, a pragmatic, multinational, randomized, controlled trial of 1360 patients with kidney failure tested the effects of high-dose HDF compared with standard hemodialysis.[199] The authors observed a reduction in death from any cause in the HDF group compared with the hemodialysis group (hazard ratio, 0.77; 95% confidence interval, 0.65–0.93). However, the mechanisms of the benefit are speculative with the removal of substances with a larger molecular weight and configuration likely only one component of the effect. Moreover, the impact on the more nebulous uremic syndrome and uremic symptomology is not clear. Future studies will undoubtedly shed more light on this.

Since the membrane pore size and selectivity are the key to removing middle molecules, a new dialysis membrane, the medium cut-off (MCO) dialyzer, has been designed to expand the efficient clearance of large middle molecules and minimize serum albumin loss. Preliminary data based on small studies suggested that MCO dialysis increased the clearance of a wide range of large middle molecules, such as free light chains, complement factor D, and α1-microglobulin, without significant risk of hypoalbuminemia.[200] Observational studies suggest that the expanded clearance of large middle molecules using MCO dialysis associated with higher health-related quality of life (HRQOL) scores and reduced symptoms such as restless legs syndrome.[201] Larger randomized clinical trials are being conducted to assess the impact of MCO dialysis on patient symptoms and long-term clinical outcomes.

## PROTEIN-BOUND SOLUTES

Other solutes that are poorly removed by standard hemodialysis include those that bind to albumin.[123,202] Their dialytic clearance is low, not because they are large molecules but because only the free, unbound solute concentration contributes to the gradient-driving solute across the dialysis membrane. In the normal kidney, the combination of protein binding and tubular secretion allows molecules to be excreted while keeping their concentrations in the ECF low.[8] This presumably represents an evolutionary adaptation to excrete toxic substances, and there is indeed suggestive evidence that some important uremic toxins are protein bound.[203] The clearance of protein-bound solutes can be increased by raising dialysate flow and membrane pore size above the levels used in conventional hemodialysis or by combining a high hemofiltration rate with dialysis in hemodiafiltration treatment.[204,205] The aggregate toxicity of protein-bound solutes could thus theoretically be assessed by comparing the effects of different renal replacement prescriptions, but this has not been attempted to date.

Studies carried out raise the possibility that levels of protein-bound solutes, like levels of large solutes, may not decline in proportion to the increase in clearance during treatment.[197] Peritoneal dialysis clears protein-bound solutes at a low rate, and the total clearance of protein-bound solutes in patients maintained on peritoneal dialysis therefore depends heavily on the level of residual kidney function.[208,209] Surprisingly, however, plasma concentrations of the bound solutes *p*-cresol sulfate and indoxyl sulfate are not much higher in patients undergoing peritoneal dialysis without residual kidney function than in those with residual kidney function, suggesting that production of these solutes may diminish as residual kidney function is lost.[209] The HEMO study assessed the effect of a higher dialysis dose (target equilibrated Kt/V 1.45) and a high-flux dialyzer membrane on all-cause mortality in patients undergoing hemodialysis three times per week compared with standard dose (target equilibrated Kt/V 1.05) and a low-flux membrane.[192,210] The Frequent Hemodialysis Network (FHN) trial aimed to determine whether increasing the frequency of in-center hemodialysis would result in beneficial changes in left ventricular mass, self-reported physical health, and other intermediate outcomes among patients undergoing maintenance hemodialysis.[192,210] Plasma concentration of *p*-cresol sulfate and indoxyl sulfate did not fall in proportion to the increase in time-averaged clearance in the more aggressively dialyzed patient groups in either the HEMO or FHN studies.[192,210]

Other early-stage studies have used a competitor substance to displace solutes from their binding sites on albumin, as well as used changes to the physical conditions within the dialyzer to increase the free fraction of protein-bound solutes (e.g., modifying osmotic conditions, temperature, or pH within the dialysis circuit).[206,207] Preventative strategies targeting intestinal microbiota and colon-derived uremic solutes, such as indoxyl sulfate (IS) and p-cresol sulfate (PCS), have also seen increased focus. For example, interventions including prebiotics, probiotics, and oral adsorbents have been studied.[206]

## SEQUESTERED SOLUTES

Some solutes are sequestered or held in compartments where their concentration does not equilibrate rapidly with that of the plasma.[211] Application of a high dialytic clearance may rapidly lower the plasma concentration of such solutes while removing only a small portion of the total body content. When this happens, intermittent dialysis treatment will be followed by a rebound in the plasma solute concentration toward predialysis levels.[86,212] Theoretically, the contribution of sequestered solutes to uremic toxicity, like the contribution of large solutes or protein-bound solutes, could be assessed by comparing the efficacy of different dialysis prescriptions. When treatment is intermittent, the removal of sequestered relative to freely equilibrating solutes can be increased by lengthening the treatment time while simultaneously reducing the plasma clearance. To date, however, while prolonged treatment has been shown to lower plasma levels of phosphate, its effect on levels of organic uremic solutes has not been observed.[180,190]

## EFFECTS OF DIET AND GASTROINTESTINAL FUNCTION

It may be possible to identify uremic toxins by comparing the effects of different diets, as well as by comparing the effects of different renal replacement therapies. Patients with kidney failure tend to reduce their intake of protein spontaneously.[213] Before dialysis was available, physicians found that protein restriction could ameliorate uremic symptoms.[214] These findings suggest that important uremic toxins are derived from protein catabolism. They call into question current recommendations that patients undergoing dialysis ingest a higher protein intake than what has been recommended for the general population.[215,216] Uremic solutes whose production depends on protein intake include urea, MG, GSA, and the indoles and phenols produced by the action of gut bacteria on tryptophan, phenylalanine, and tyrosine.[95,107,217-219] This group overlaps with the large group of uremic solutes made by colon microbes.[155] The production of such solutes may depend on not only dietary intake but also gut function. Impaired small bowel function may increase the delivery of peptides to the colon in uremia, and the composition of the colon microbiome may also be altered.[220,221] If colonic bacteria produce uremic toxins, uremic symptoms could theoretically be relieved by altering the delivery of substrates to the colon or adding sorbents to the diet. Only limited studies of such maneuvers have so far been performed, with success most often obtained with the use of dietary fiber or in altering the protein quantity (shifting to low) and type (shifting to plant-based protein) to reduce the production of selected colon-derived solutes.[222-225] Phosphate binders and novel sorbent agents have also shown some promise in modulating specific gut-derived putative toxins.[225] Historically, once hemodialysis became widely available, attempts to modify uremic solute production were largely abandoned. However, interest in this area has been revived by the imperfect efficacy of conventional dialysis and by the relatively disappointing results to date of trials evaluating more intensive dialysis treatment.[119] Interest has also been stimulated by new knowledge of the microbiome and the promise that if toxic solutes were better known, the microbiome composition or enzymatic function could be manipulated to reduce their production.[226]

Indeed, significant carbamylation reductions (measured by homocitrulline levels in the blood) have been achieved through dietary interventions, employing either a very low protein diet supplemented with keto-analogues or a Mediterranean diet, both compared to a free diet.[227] In a randomized, crossover, controlled trial, 60 patients with advanced CKD were assigned to each diet (very low protein diet supplemented with keto-analogues, Mediterranean diet, or free diet). Compared with a free diet, both the very low protein and Mediterranean diets were associated with decreases in serum homocitrulline levels, and such reduction correlated with urea reduction.

## SOLUTE CLEARANCE BY ORGANIC TRANSPORT SYSTEMS

The cloning of transporters, which move organic solutes into the lumen of the proximal tubule, has provided a potential new means to identify potential uremic toxins.

To the extent that uremia is caused by accumulation of organic solutes, knocking out these transporters would be expected to recapitulate uremic symptoms. To date, knocking out individual transporters has been found not to cause detectable illness, likely because of redundancy of the transport systems.[228,229] The accumulation of uremic solutes may interfere with organic solute transport important for detoxification at other sites, most notably the liver and blood-brain barrier.[230-232]

## METABOLIC EFFECTS OF UREMIA

The loss of kidney function has numerous metabolic effects. Some of the most prominent are listed in Box 51.1. A few can be related to the loss of specific renal processes, such as the hydroxylation of vitamin D. However, most have no clear cause and can at present only be attributed to the retention of uremic solutes.

### OXIDANT STRESS AND THE MODIFICATION OF PROTEIN STRUCTURE

Studies have suggested that loss of kidney function increases oxidant stress.[233] The term "oxidant stress" is acknowledged to be vague, although a wealth of evidence points to increased oxidant effects in uremia. Increased levels of primary oxidants cannot be documented because they are evanescent species, which act locally, such as superoxide anion, hydrogen peroxide, hydroxyl radical, and hypochlorous acid. The accumulation of various by-products of oxidant reactions is therefore taken as evidence of increased oxidant activity. Although the accumulation of these markers of oxidant activity is well documented, there is at present no explanation as to why the production of oxidants should be increased in uremia. Leukocyte activation leading to increased production of hypochlorous acid has been described in patients undergoing dialysis and may be especially prominent when uremia is accompanied by systemic inflammation.[234]

Among the most commonly measured markers of oxidant activity have been oxidized amino acids and related compounds and substances that react with thiobarbituric acid including malondialdehyde.[235] The accumulation of these low-molecular-weight compounds could reflect reduced renal clearance, as well as increased production. More convincing evidence of oxidant stress is the accumulation of intact proteins containing oxidized amino acids.[236,237] The accumulation of these larger markers of oxidation cannot be attributed to reduced renal clearance. Further potential evidence of oxidative stress in uremia is the loss of extracellular reducing substances. The extracellular compartment is normally provided with several reducing substances, of which the reduced forms of ascorbic acid and plasma albumin are considered to be the most important. In uremia, the portion of ascorbic acid and albumin circulating in the oxidized form is increased. The case of albumin, which undergoes oxidation at its single free cysteine thiol (SH) group, is particularly interesting. Plasma albumin in patients with kidney failure is rapidly restored to the reduced form during hemodialysis.[238] The shift to oxidized albumin in untreated uremia is associated with the accumulation of cystine, which is the oxidized form of the thiol amino acid cysteine, and the shift back to reduced albumin during hemodialysis is associated with a lowering of cystine levels toward normal. One explanation for these phenomena is that normal kidney function is required to accomplish the steady reduction of cystine and albumin, which must take place to offset normal oxidant production.

The major ill effect of increased oxidant activity in uremia is thought to be modification of proteins. Proteins are modified by not only direct oxidation of amino acids but also the combination of amino acid side chains with carbonyl (C = O) compounds. The terminology in this area is confusing. The first carbonyl compounds shown to react with proteins were sugars, and the modified proteins formed after several reaction steps were therefore referred to as advanced glycosylation end products (AGEs). Elevated sugar concentrations could account for the increased concentrations of AGEs found in persons with diabetes mellitus but not for the subsequent findings of similarly increased AGE concentrations in patients with ESKF. Studies have shown that high levels of active nonsugar carbonyls are responsible for the increased production of these modified proteins when renal function is reduced.[239] The active carbonyls have not been fully characterized, but they include compounds such as glyoxal (see Fig. 51.1), which can be produced by oxidation of sugars and lipids. It has therefore been suggested that the protein end products of carbonyl modification in uremia should be referred to not as AGEs but as advanced glycoxidation and lipoxidation end products. Terminology aside, interest in both directly oxidized and carbonyl-modified proteins has centered on the possibility that alterations in protein structure contribute to uremia.[240,241] The hypothesis that oxidant stress contributes to adverse health consequences in ESKF has prompted trials of various antioxidants. So far, administration of vitamin C, various forms of vitamin E, folate, and α-lipoic acid has failed to reverse plasma indices of oxidant stress in patients undergoing dialysis.[242-247]

### EFFECTS OF UREMIA

#### RESTING ENERGY EXPENDITURE

Resting energy expenditure has been reported as increased, decreased, and normal in patients with kidney failure.[248-252] The choice of control populations and other methodologic issues, such as corrections for altered body composition, have probably contributed to this uncertainty. The effect of dialysis treatment, which may transiently increase energy expenditure, must also be considered. Uremia, apart from dialysis, likely reduces resting energy expenditure.[250,252] Lower energy expenditure accords with observations of lower body temperatures in those with uremia, although additional factors may be at play in thermoregulation.[17] However, dialysis treatment may itself speed metabolism and increase energy expenditure.[251] Effects of inflammation in patients with ESKF add further complexity to the assessment of energy requirements.[253] Using anthropometric techniques in patients with ESKF on chronic hemodialysis who were maintained for 3 months on carefully constructed diets, the energy requirements were found not too different from those reported for similarly sedentary normal subjects.[254]

The energy requirements in ESKF were, however, more variable than in normal subjects.

Knowledge of the physiologic control mechanisms for appetite, fat metabolism, and energy expenditure is potentially relevant to uremic energy metabolism. Studies have focused on the signaling molecules ghrelin, produced by the stomach, and leptin, produced by adipose tissue along with other adipose tissue–derived hormones (adipokines). Levels of these small proteins tend to rise in kidney failure because of reduced clearance by the kidney and possibly because of increased production.[74,255]

## CARBOHYDRATE METABOLISM

Insulin resistance is the most conspicuous derangement in uremic carbohydrate metabolism.[256] The defect is clearly present in ESKF, but in cross-sectional studies, impairment can be detected when the GFR falls below 50 mL/min/1.73 m,[5] with a graded relation to GFR.[257,258] There are probably several causes of this phenomenon.[259] Surprisingly, some obvious possibilities do not seem to contribute. Insulin binds normally to its receptor in uremia, and receptor density is unchanged.[260,261] Moreover, excess levels of glucagon or fatty acids do not account for the disorder.[262] As is the case with overall energy metabolism, interest has recently focused on the contribution of adipokines including leptin, resistin, and adiponectin to insulin resistance. Adipose tissue has also been identified as a source of inflammatory cytokines that impair insulin action in various experimental systems and circulate at increased levels in many patients with advanced renal insufficiency. However, correlations among levels of individual substances with measures of insulin resistance are poor, and the extent to which adipose tissue products contribute to uremic insulin resistance remains uncertain.[75,262,263]

Because dialysis, transplantation, and low protein diets tend to restore insulin responsiveness, it has also been suggested that unidentified nitrogenous product(s) mediate insulin resistance.[262] Acidosis has been shown to provoke insulin resistance and accumulation of acid, as well as nitrogenous wastes, may contribute to insulin resistance in uremia.[264] 11-β-Hydroxysteroid dehydrogenase type 1 provokes insulin resistance by regenerating glucocorticoids. In experimental uremia, liver and fat tissues express increased activity of this enzyme, as well as insulin resistance. The insulin resistance associated with 11-β-hydroxysteroid dehydrogenase can be mitigated with an inhibitor of the enzyme, all suggesting a role for this steroid pathway in insulin resistance.[265] The cause for increased enzymatic activity is unknown. Resistin is a protein capable of inducing insulin resistance, and its levels are high when kidney function is impaired. However, the plasma concentrations of resistin are not associated with insulin resistance if the GFR is taken into account.[75] Retinol-binding protein is also associated with insulin resistance (and also rises in ESKF) but is not related to markers of glucose control.[266] Finally, physical inactivity and deconditioning may contribute to insulin resistance. Exercise programs have been shown to mitigate insulin resistance but must be relatively protracted and intensive to be effective.[267,268] Ongoing studies have shown exercise programs in CKD and ESKF to yield various benefits including improved physical strength and improved cardiovascular and kidney health measures. However, the mechanisms for these benefits independent of those found in the general population remain speculative and the impact specifically on uremic toxins as mediators is still unanswered.[269,270] Improvements in insulin sensitivity and reductions in systemic inflammation and oxidative stress are leading mechanistic contenders.[270]

Insulin resistance may have several adverse effects. Most importantly, it has been recognized as a risk factor for cardiovascular disease.[271] The connections between insulin resistance and vascular disease are not clear. A tendency to hyperglycemia is one presumably toxic effect. Some investigators have suggested that the sodium-retentive effect of insulin on the kidney remains intact, whereas other tissues become insulin resistant in uremia. Increased plasma insulin concentrations could thus contribute to arterial hypertension in patients with impaired kidney function.[272,273] Outside of vascular disease, the loss of insulin's anabolic action may contribute to uremic muscle wasting.[262,274]

Even though insulin resistance is the rule in uremia, hypoglycemia can be a significant effect of renal insufficiency.[275] Hypoglycemia is likely to occur, despite insulin resistance, for two main reasons. First, the kidney is a major site of insulin catabolism. Patients with diabetes mellitus treated with insulin, or insulin secretagogues (e.g., sulfonylureas), frequently become hypoglycemic if doses are not adjusted downward as the GFR declines. Second, the kidney is a major site of gluconeogenesis.[275] The liver produces the bulk of glucose in postabsorptive and starvation states, but even in these situations, the kidney produces some glucose. With prolonged fasting, the kidney is responsible for approximately half of the total glucose production.[276,277] Thus advanced CKD may predispose to hypoglycemia by prolonging insulin action and reducing gluconeogenesis. These effects may become particularly apparent when other hypoglycemic factors, such as ethanol ingestion or liver disease, are also evident.

## LIPID METABOLISM

Nephrotic syndrome and even lower-grade proteinuria are regularly associated with hyperlipidemia.[278] However, lipid abnormalities are modest when kidney function is impaired without significant proteinuria.[279] Indeed, total cholesterol falls on average as the GFR drops below about 30 mL/min/1.73 $m^2$.[213] Metabolite profiling in plasma of patients with ESKF using liquid chromatography and tandem mass spectrometry has revealed that lipid products deviate from normal levels far less than polar compounds. However, lower-molecular-weight triacylglycerols were generally decreased, and an increase in intermediate-weight triacylglycerols was observed.[6] The causes and consequences of these changes in lipids are uncertain. The decline in total cholesterol is taken to reflect, at least in part, progressive reduction in food intake. Numerous hypotheses have been advanced to account for the finding that atherosclerosis is accelerated, but lipid levels are not markedly elevated in renal failure. Even though total lipid levels are not significantly elevated, there may be an increase in oxidized forms due to oxidant stress and reduced lipoprotein clearance rates.[280] The high prevalence of cardiovascular disease in patients with ESKF has prompted several randomized clinical trials of statin agents. The largest trials performed to date have identified no clear benefit of statins in patients on dialysis, in sharp contrast to persons with normal or near-normal kidney function and, in one study,[281] to patients with advanced, non–dialysis-requiring CKD.[282-284] Furthermore, statins seem to provide no benefit in slowing the progression of kidney disease.[285] Nevertheless, given the high cardiovascular risks in patients with nondialysis CKD, most recommend statin therapy as secondary prevention for nondialysis CKD patients with atherosclerotic cardiovascular disease (similar to those without CKD) and even for primary prevention in those older than the age of 50 or at high risk.[286] For ESKF patients on dialysis, the evidence suggests not routinely initiating statin therapy but maintaining it if already receiving it at the time of dialysis initiation.[287]

### AMINO ACID AND PROTEIN METABOLISM

The normal kidney participates in the metabolism of several amino acids.[11,288-291] For example, the kidney converts citrulline to arginine, one of the necessary steps in the urea cycle. Loss of this function likely contributes to the increasing ratio of citrulline to arginine as the GFR declines below 50 mL/min/1.73 $m^2$.[289,290] Similarly, reduced renal production of serine from glycine probably underlies the rise in the plasma glycine-to-serine ratio. Increased concentrations of the sulfur-containing amino acids—cysteine, taurine, and homocysteine—are especially intriguing. Cysteine and homocysteine accumulate in the oxidized form, consistent with the concept that uremia is a state of oxidant stress, and homocysteine levels have been associated with the progression of cardiovascular disease.[289,290,292] Administration of folate to lower homocysteine levels neither restores the plasma redox state toward normal nor reduces the frequency of cardiovascular events.[244,293] The mechanism responsible for the accumulation of oxidized cystine and homocysteine remains unclear, but this change can be detected as the GFR drops below roughly half of normal and becomes more extreme as ESKF approaches.

Tissue protein loss reflected by muscle wasting is a major concern in patients with kidney failure. Factors that predispose to protein wasting include reduced appetite, along with insulin resistance and altered amino acid metabolism, as described earlier. Dialysis also results in some protein loss, with amino acids lost in the dialysate and plasma proteins and amino acids lost in the peritoneal dialysate. In the absence of other complications, the effect of uremia at the levels now seen clinically on protein metabolism is usually modest, at least over the short term.[274] Patients with advanced CKD can maintain nitrogen balance on low protein diets as long as acidosis and inflammation are minimal. Several factors may combine with defects in insulin resistance and altered amino acid and adipose metabolism to produce muscle wasting. The best studied of these is acidosis, which has been shown to stimulate the ubiquitin–proteasome pathway of intracellular protein degradation. Activation of caspase-3 seems to be an important step in proteolysis, which is followed by disposal of protein cleavage fragments through the proteasome.[294] In addition to these effects, acidosis contributes to insulin resistance and thereby attenuates protein anabolic actions of insulin.[295] Base supplements can mitigate the catabolic effects of acidosis, but a long-term study establishing the value of normalizing bicarbonate levels in patients with impaired kidney function is lacking.[264,296-298] Newer guidelines no longer recommend a target serum bicarbonate level or a graded treatment recommendation. Instead, the guidelines simply suggest treating to keep serum bicarbonate levels >18 mEq/L.[299]

Inflammation may be an even more important contributor to protein wasting than acidosis in patients with kidney failure. Muscle loss in patients with ESKF has been linked to an inflammatory state, characterized by increased serum concentrations of C-reactive protein and various cytokines. How these inflammatory mediators trigger net protein degradation in muscle and other tissues remains to be better elucidated, although their presence is regularly accompanied by muscle loss and theoretical pathways are proposed.[300] High levels of inflammatory mediators are also associated with lower serum albumin concentrations, which have been attributed largely to reduced hepatic production of albumin. Both muscle loss and reduced albumin concentration predict early death. The exact cause(s) of inflammation remains elusive. In some cases, inflammation can be ascribed to known episodes of infection or other intercurrent illness, and in other cases, periodontal disease and gut dysbiosis can be invoked.[300] In many cases, however, no cause can be identified. Occult inflammatory stimuli in these cases may include subclinical infection at hemodialysis catheter or arteriovenous graft sites, exposure to dialysate and various synthetic materials, and accelerated vascular injury that is common in ESKF.[301-303] Oxidant stress has also often been invoked. Attempts to reduce inflammation with free radical scavengers and other agents, however, have been largely unsuccessful.[246,304,305] An interesting possibility is that organic solutes retained in kidney failure, although they do not regularly trigger inflammation over the short term, cause the late appearance of inflammation in a subset of patients with ESKF. Some evidence has suggested that accumulation of proteins modified by glycation and oxidation can trigger a self-perpetuating inflammatory loop in these cases.[306,307]

Another factor contributing to muscle wasting is inactivity. Johansen and associates[308] have shown that self-reported physical activity in patients starting dialysis is at, or below, the first percentile for population reference range, and lower levels of physical activity are strongly associated with mortality.[309] In these patients, inactivity may be caused by fatigue and loss of energy, which are invariable although difficult-to-measure features of uremia, and by depression and other comorbid illnesses,[310] which are common in the ESKF population. Exercise and other physical modality interventions in CKD have shown mixed but promising results through a variety of proposed mechanisms.[311-313]

## OVERALL NUTRITION

As emphasized by Depner,[13] the condition of patients undergoing dialysis reflects a combination of residual uremia, side effects of dialysis mixed with the effects of comorbid conditions, and increasing age. Most patients starting on dialysis in Europe and the United States are overweight. This reflects population-wide overeating, which can contribute to and accelerate the progression of CKD. The protein wasting exhibited by a subset of patients undergoing dialysis is thus not malnutrition in the sense of limited nutrient availability. It is often accompanied by anorexia and reduced food intake, particularly when inflammation is prominent. It cannot, however, be reversed simply by increasing food intake.[314] Other measures of restoring body protein and muscle mass toward normal, including exercise, appetite stimulants, and newer anabolic and antiinflammatory agents, are under study.

## SIGNS AND SYMPTOMS OF UREMIA

Frequently identified signs and symptoms of uremia are listed in Box 51.1. That a fundamental metabolic disturbance such as uremia should have such a wide variety of consequences is not remarkable. The complications of untreated diabetes or hyperthyroidism are similarly extensive. However, uremia is different in that we cannot trace all its complications to dysregulation of a single key compound. And, except for renal transplantation, current therapy for uremia cannot return patients as close to normal as thyroid hormone or insulin replacement. Further highlighting the complexity of targeting compounds to ameliorate the signs and symptoms of uremia is that uremic symptoms don't even necessarily track tightly with reductions in eGFR.[315]

The level of lost kidney function at which uremia can be said to appear is obscure. Furthermore, the diminution of functions other than solute clearance likely contributes to the symptoms and signs of uremia. In general, these other functions, such as ammoniagenesis, erythropoietin and 1,25-dihydroxyvitamin D synthesis, urine-concentrating capacity, and tubular secretion, tend to decline in parallel with GFR, but not always. Nevertheless, defining the level of kidney function solely by GFR may be misleading. For example, certain potentially toxic solutes depend more on tubular secretion than on glomerular filtration for their excretion, and renal synthetic processes are probably linked to GFR only by virtue of the loss of functioning renal tissue. However, until particular kidney dysfunctions are attached to specific aspects of the uremic syndrome, GFR will remain the principal index of kidney function.

Most of the clinical and biochemical characteristics of uremia have been defined in ESKF or at a level of GFR near ESKF, but recent work shows symptoms can arise much earlier in CKD.[315] Furthermore, uremic characteristics may be hard to dissect from complications of the dialysis procedure itself in ESKF. Other morbidities considered separate from uremia also commonly interact with it and can contribute to symptoms. For example, the cardiovascular disease suffered especially by patients with diabetes and hypertension appears to be accelerated by CKD. However, the myocardial infarctions, strokes, and peripheral vascular diseases suffered by these patients have not traditionally been considered features of the uremic syndrome. These conditions nevertheless add to patients' disabilities in ways that are often not easily distinguishable from uremia or the residual syndrome of ESKD. Similarly, the peripheral neuropathy and gastroparesis of diabetes are difficult to disentangle from uremic neuropathy and uremic anorexia, nausea, and vomiting.

## WELL-BEING AND PHYSICAL FUNCTION

The point in the course of CKD at which quality of life begins to decline has not been dissected in great detail, but some data exist showing limited correlations with eGFR.[315] Given the list of signs and symptoms in Box 51.1, it is not surprising that HRQOL tends to decline in patients with CKD.[316-319] Dialysis undoubtedly imposes a burden on patients. Interestingly, however, comparisons of HRQOL in patients on dialysis and patients with advanced CKD who are not on dialysis have yielded discordant results.[320-326] Determining predictors of symptom trajectory—those whose symptoms improve with dialysis versus those whose symptoms do not—could have great impact on care planning for patients. Other, often neglected features of treatment, such as pill burden and frailty, may also contribute to reduction in the quality of life.[327] Patients with ESKD are on average more depressed than healthy controls. However, it is difficult to distinguish the extent to which depression is caused by uremic solutes as compared with the effects of comorbid disease and the knowledge of ill health and limited life expectancy. Not surprisingly, transplantation has been found to consistently improve quality of life.[328] However, early versus later initiation of chronic hemodialysis does not seem to influence quality of life.[329]

Physical functioning in patients treated with dialysis is decidedly below normal.[330] The self-reported activity of people initiating dialysis is below the fifth percentile for healthy people.[308] Treatment of anemia improves this situation but does not normalize it.[331,332] The most detailed studies have identified multiple defects that are associated with fatigability.[333] These include muscle energetic failure and neural defects. The degree to which they are attributable to the uremic environment itself, deconditioning, and/or comorbid conditions such as diabetes mellitus is difficult to establish. Even highly functional patients on dialysis display notable physical limitations: Balance, walking, speed, and sensory function in dialysis patients were clearly below those of matched controls, and incident dialysis patients showed decline over 2 years.[334,335]

## NEUROLOGIC FUNCTION

A particularly interesting group of uremic signs and systems reflects altered nerve function. Classic descriptions emphasized that uremic patients could appear alert, despite defects in memory, planning, and attention.[17,336] As kidney function worsened, patients progressed to coma or catatonia, which could be relieved by dialysis. Today, patients maintained on dialysis exhibit more subtle cognitive defects.[337] A difficulty in identifying the effects of uremia in these patients is that the hemodialysis procedure and/or associated factors (e.g., hypotension) may transiently impair cognitive function.[338] Studies in patients with CKD have suggested that cognitive impairment can be detected when the GFR falls below 60 mL/min/1.73 $m^2$ and worsens as the GFR declines.[339-341] As with other signs and symptoms of uremia, the degree to which cognition is influenced by uremia, as opposed to other comorbidities, especially cerebrovascular disease, is difficult to ascertain. Imaging studies suggest that subclinical cerebrovascular disease is common in CKD, and its role in poor cognition needs further definition.[342-344] The finding that kidney transplantation improves cognitive function suggests, however, that at least some of the impairment observed in ESKF patients is reversible and, perhaps, due to solute accumulation.[345-347] Impaired cognitive function has so far been associated with accumulation of one specific uremic solute, 4-hydroxyphenylacetate, and has been shown to be partly reversed by acute dialysis treatment.[133,348] A further reflection of altered CNS function in uremia is impaired sleep.[349,350] Sleep is fragmented by brief arousals and apneic episodes, which are often associated with bursts of repetitive leg movement or restless legs syndrome.[351,352]

Sensorimotor neuropathy was a recognized component of the uremic syndrome decades ago.[17] Studies of conduction velocity and other nerve functions have since repeatedly found that most patients with uremia have peripheral neuropathy, albeit often subclinical.[336,353,354] Morphologic studies have shown that these functional changes are associated with axonal loss and therefore may not be reversible in some patients. The extent to which peripheral nerve function is impaired earlier in the course of CKD is not certain. Autonomic neuropathy also develops in ESKF but has been less extensively studied than peripheral neuropathy.[354] As with other uremic disturbances, the cause of neuropathy is unknown. Parathyroid hormone and the retention of multiple solutes, especially potassium, have been associated with peripheral neuropathy without definitive proof of causality.[336,353]

## PRURITIS

Pruritus among patients with CKD is associated with lower HRQOL and poor sleep, and it is an independent predictor of mortality.[355-357] Despite an estimated mean prevalence of 41% in patients on dialysis and the emergence of novel treatments, pruritus is often underrecognized by dialysis providers and undertreated.[358-360] Though often thought of as a symptom of advanced CKD approaching dialysis, the prevalence of pruritus in patients with eGFR between 30 and 60 mL/min/1.73 m$^2$ may approach or exceed 40%.[361] The exact pathophysiology of kidney disease–associated pruritus associated with kidney disease remains unclear, with theories regarding the underlying biologic mechanism of pruritus primarily including as of yet unidentified uremic toxins, dysregulation of mineral metabolism, and altered immunochemical pathways.[362] Because the majority of kidney-disease associated pruritus research has been done in the dialysis population, it is also unknown whether the pathophysiology of pruritus in nondialysis CKD shares a common mechanism.

## APPETITE, TASTE, AND SMELL

Loss of appetite is a common uremic symptom and presumably contributes to malnutrition in patients with advanced renal failure. A large number of causes have been proposed. Acidosis and inflammatory cytokines, including tumor necrosis factor and various interleukins, have been identified as contributing factors.[363] As with the uremic defects in energy metabolism, attention has been focused on the accumulation of small proteins produced by the gut and adipose tissue and acting on the brain to regulate appetite in normal people.[364,365] Levels of leptin, an anorexigen produced by adipose tissue, are elevated in ESKD. Antagonism of leptin in mice with experimental CKD attenuates a number of the molecular markers of proteolysis.[366] An interesting feature of uremic anorexia that remains to be explained is a disproportionate reduction in the intake of protein.[213] Along with overall loss of appetite, erosion of taste and smell has long been recognized in the CKD and ESKF populations.[367-370] As with most defects, transplantation reverses the blunted smell.[367] Studies have demonstrated that odor identification was impaired for both food and nonfood substances in >90% of dialysis-dependent ESKF patients (~65% had moderate, severe, or total deficits; many were unaware of deficits on subjective assessment).[370] Importantly, lower olfactory identification scores were associated with worse nutritional status—indicated by higher subjective global assessment scores (a validated score to assess food intake, weight changes, and symptoms such as nausea and anorexia) and lower levels of serum albumin, total cholesterol, and LDL cholesterol. Taste acuity has been reported as lower in patients undergoing dialysis than in those with renal insufficiency, and self-reported altered taste is associated with poor nutritional status.[371,372] The factors responsible for these defects are again unknown.

## CELLULAR FUNCTIONS

The most general cellular abnormality reported has been the inhibition of sodium–potassium adenosine triphosphatase ($Na^+$, $K^+$-ATPase). Decreased $Na^+$, $K^+$-ATPase activity in red cells of uremic patients was reported in 1964.[373] In general, subsequent reports have confirmed the observation, noted the same effect in other cell types, and emphasized that the inhibition was attributable to some factor in uremic serum.[374] The evidence for a circulating inhibitor includes the findings that dialysis reduces the inhibitory activity and uremic plasma can acutely suppress the pump activity.[374] However, the factor or factors have remained elusive. A number of candidates have been considered. Much attention has focused on digitalis-like substances. Several such compounds have been found in excess in humans with ESKF. These include marinobufagenin and telocinobufagin, which have a structure related to that of digitalis.

# CONCLUSIONS

Uremia is considered a clinical syndrome that can affect numerous organs and systems in the human body. Common manifestations include loss of appetite, altered smell and taste, nausea, vomiting, progressive weakness and fatigue, neuropathy, impaired sleep, altered mentation, and pruritus. In addition to symptoms, several metabolic and cellular alterations are noted with uremia and likely contribute to adverse outcomes in patients with kidney disease. Dialysis can reduce but seldom completely resolves uremic signs and symptoms in most patients. Still, observations suggest that retained solutes are the primary cause of uremia and the list of putative uremic toxins is ever expanding due to modern analytic techniques. However, proving causality is difficult in large part due to the multiplicity of solutes and signs and symptoms associated with reduced kidney function. Better understanding of which solutes are toxic and their mechanisms of action is required to significantly advance therapeutic options for uremia.

*Visit Elsevier eBooks+ (eBooks.Health.Elsevier.com) for a complete set of references.*

## KEY REFERENCES


4. Vanholder R, De Smet R, Glorieux G, et al. Review on uremic toxins: classification, concentration, and interindividual variability. *Kidney Int*. 2003;63:1934–1943.
5. Duranton F, Cohen G, De Smet R, et al. Normal and pathologic concentrations of uremic toxins. *J Am Soc Nephrol*. 2012;23:1258–1270.
8. Sirich TL, Aronov PA, Plummer NS, Hostetter TH, Meyer TW. Numerous protein-bound solutes are cleared by the kidney with high efficiency. *Kidney Int*. 2013;84:585–590.
13. Depner TA. Uremic toxicity: urea and beyond. *Semin Dial*. 2001;14:246–251.
14. Johnson WJ, Hagge WW, Wagoner RD, Dinapoli RP, Rosevear JW. Effects of urea loading in patients with far-advanced renal failure. *Mayo Clin Proc*. 1972;47:21–29.
22. Wang Z, Nicholls SJ, Rodriguez ER, et al. Protein carbamylation links inflammation, smoking, uremia and atherogenesis. *Nat Med*. 2007;13:1176–1184.
41. Berg AH, Drechsler C, Wenger J, et al. Carbamylation of serum albumin as a risk factor for mortality in patients with kidney failure. *Sci Transl Med*. 2013;5:175ra29.
42. Kalim S, Trottier CA, Wenger JB, et al. Longitudinal changes in protein carbamylation and mortality risk after initiation of hemodialysis. *Clin J Am Soc Nephrol*. 2016;11:1809–1816.
43. Kalim S, Zhao S, Tang M, et al. Protein carbamylation and the risk of ESKD in patients with CKD. *J Am Soc Nephrol*. 2023;34:876–885.
60. Verroust PJ, Birn H, Nielsen R, Kozyraki R, Christensen EI. The tandem endocytic receptors megalin and cubilin are important proteins in renal pathology. *Kidney Int*. 2002;62:745–756.
63. Norden AG, Sharratt P, Cutillas PR, Cramer R, Gardner SC, Unwin RJ. Quantitative amino acid and proteomic analysis: very low excretion of polypeptides >750 Da in normal urine. *Kidney Int*. 2004;66:1994–2003.
89. Wyss M, Kaddurah-Daouk R. Creatine and creatinine metabolism. *Physiol Rev*. 2000;80:1107–1213.
116. Cathcart-Rake W, Porter R, Whittier F, Stein P, Carey M, Grantham J. Effect of diet on serum accumulation and renal excretion of aryl acids and secretory activity in normal and uremic man. *Am J Clin Nutr*. 1975;28:1110–1115.
117. Mitch WE, Brusilow S. Benzoate-induced changes in glycine and urea metabolism in patients with chronic renal failure. *J Pharmacol Exp Ther*. 1982;222:572–575.
119. Meyer TW, Hostetter TH. Uremic solutes from colon microbes. *Kidney Int*. 2012;81:949–954.
124. Sirich TL, Luo FJ, Plummer NS, Hostetter TH, Meyer TW. Selectively increasing the clearance of protein-bound uremic solutes. *Nephrol Dial Transplant*. 2012;27:1574–1579.
127. Vanholder R, Schepers E, Pletinck A, Nagler EV, Glorieux G. The uremic toxicity of indoxyl sulfate and p-cresyl sulfate: a systematic review. *J Am Soc Nephrol*. 2014;25:1897–1907.
138. Arinze NV, Yin W, Lotfollahzadeh S, et al. Tryptophan metabolites suppress the Wnt pathway and promote adverse limb events in chronic kidney disease. *J Clin Investig*. 2022;132:e142260.
148. Ponda MP, Quan Z, Melamed ML, Raff A, Meyer TW, Hostetter TH. Methylamine clearance by haemodialysis is low. *Nephrol Dial Transplant*. 2010;25:1608–1613.
162. Melamed ML, Plantinga L, Shafi T, et al. Retained organic solutes, patient characteristics and all-cause and cardiovascular mortality in hemodialysis: results from the retained organic solutes and clinical outcomes (ROSCO) investigators. *BMC Nephrol*. 2013;14:134.
163. Simenhoff ML, Burke JF, Saukkonen JJ, Ordinario AT, Doty R. Biochemical profile or uremic breath. *N Engl J Med*. 1977;297:132–135.
165. Tang WH, Wang Z, Levison BS, et al. Intestinal microbial metabolism of phosphatidylcholine and cardiovascular risk. *N Engl J Med*. 2013;368:1575–1584.
168. Stubbs JR, House JA, Ocque AJ, et al. Serum trimethylamine-N-oxide is elevated in CKD and correlates with coronary atherosclerosis burden. *J Am Soc Nephrol*. 2016;27:305–313.
169. Shafi T, Powe NR, Meyer TW, et al. Trimethylamine N-oxide and cardiovascular events in hemodialysis patients. *J Am Soc Nephrol*. 2017;28:321–331.
189. Wen D, Zheng Z, Surapaneni A, et al. Metabolite profiling of CKD progression in the chronic renal insufficiency cohort study. *JCI Insight*. 2022;7:e161696.
190. Kalim S, Wald R, Yan AT, et al. Extended duration nocturnal hemodialysis and changes in plasma metabolite profiles. *Clin J Am Soc Nephrol*. 2018;13:436–444.
191. Sirich TL, Meyer TW, Gondouin B, Brunet P, Niwa T. Protein-bound molecules: a large family with a bad character. *Semin Nephrol*. 2014;34:106–117.
192. Meyer TW, Sirich TL, Fong KD, et al. Kt/vurea and nonurea small solute levels in the hemodialysis study. *J Am Soc Nephrol*. 2016;27:3469–3478.
203. Vanholder R, Schepers E, Pletinck A, Neirynck N, Glorieux G. An update on protein-bound uremic retention solutes. *J Ren Nutr*. 2012;22:90–94.
210. Sirich TL, Fong K, Larive B, et al. Limited reduction in uremic solute concentrations with increased dialysis frequency and time in the Frequent Hemodialysis Network Daily Trial. *Kidney Int*. 2017;91:1186–1192.
229. Eraly SA, Vallon V, Vaughn DA, et al. Decreased renal organic anion secretion and plasma accumulation of endogenous organic anions in OAT1 knock-out mice. *J Biol Chem*. 2006;281:5072–5083.
233. Himmelfarb J. Uremic toxicity, oxidative stress, and hemodialysis as renal replacement therapy. *Semin Dial*. 2009;22:636–643.
246. Himmelfarb J, Ikizler TA, Ellis C, et al. Provision of antioxidant therapy in hemodialysis (PATH): a randomized clinical trial. *J Am Soc Nephrol*. 2014;25:623–633.
251. Ikizler TA, Wingard RL, Sun M, Harvell J, Parker RA, Hakim RM. Increased energy expenditure in hemodialysis patients. *J Am Soc Nephrol*. 1996;7:2646–2653.
254. Shah A, Bross R, Shapiro BB, Morrison G, Kopple JD. Dietary energy requirements in relatively healthy maintenance hemodialysis patients estimated from long-term metabolic studies. *Am J Clin Nutr*. 2016;103:757–765.
256. Spoto B, Pisano A, Zoccali C. Insulin resistance in chronic kidney disease: a systematic review. *Am J Physiol Ren Physiol*. 2016;311:F1087–F1108.
265. Chapagain A, Caton PW, Kieswich J, et al. Elevated hepatic 11beta-hydroxysteroid dehydrogenase type 1 induces insulin resistance in uremia. *Proc Natl Acad Sci U S A*. 2014;111:3817–3822.
282. Wanner C, Krane V, Marz W, et al. Atorvastatin in patients with type 2 diabetes mellitus undergoing hemodialysis. *N Engl J Med*. 2005;353:238–248.
283. Fellstrom BC, Jardine AG, Schmieder RE, et al. Rosuvastatin and cardiovascular events in patients undergoing hemodialysis. *N Engl J Med*. 2009;360:1395–1407.
293. Jamison RL, Hartigan P, Kaufman JS, et al. Effect of homocysteine lowering on mortality and vascular disease in advanced chronic kidney disease and end-stage renal disease: a randomized controlled trial. *JAMA*. 2007;298:1163–1170.
304. Rivara MB, Yeung CK, Robinson-Cohen C, et al. Effect of coenzyme Q(10) on biomarkers of oxidative stress and cardiac function in hemodialysis patients: the CoQ(10) biomarker trial. *Am J Kidney Dis*. 2017;69:389–399.
305. Nowak KL, Chonchol M, Ikizler TA, et al. IL-1 inhibition and vascular function in CKD. *J Am Soc Nephrol*. 2017;28:971–980.
306. Thornalley PJ. Advanced glycation end products in renal failure. *J Ren Nutr*. 2006;16:178–184.
316. Wulczyn K, Zhao S, Rhee E, Kalim S, Shafi T. Trajectories of uremic symptom severity and kidney function in patients with chronic kidney disease. *Clin J Am Soc Nephrol*. 2022;17:496–506.
337. O'Lone E, Connors M, Masson P, et al. Cognition in people with end-stage kidney disease treated with hemodialysis: a systematic review and meta-analysis. *Am J Kidney Dis*. 2016;67:925–935.
347. Van Sandwijk MS, Ten Berge IJ, Majoie CB, et al. Cognitive changes in chronic kidney disease and after transplantation. *Transplantation*. 2016;100:734–742.
352. Gopaluni S, Sherif M, Ahmadouk NA. Interventions for chronic kidney disease-associated restless legs syndrome. *Cochrane Database Syst Rev*. 2016;11:CD010690.
359. Fishbane S, Jamal A, Munera C, Wen W, Menzaghi F. A phase 3 trial of difelikefalin in hemodialysis patients with pruritus. *N Engl J Med*. 2020;382:222–232.
380. Koeth RA, Kalantar-Zadeh K, Wang Z, Fu X, Tang WH, Hazen SL. Protein carbamylation predicts mortality in ESKD. *J Am Soc Nephrol*. 2013;24:853–861.
383. Drechsler C, Kalim S, Wenger JB, et al. Protein carbamylation is associated with heart failure and mortality in diabetic patients with end-stage renal disease. *Kidney Int*. 2015;87:1201–1208.

## REFERENCES


1. Eknoyan GA. History of uremia research. *J Ren Nutr*. 2017;27:449–452.
2. Meyer TW, Hostetter TH. Uremia. *N Engl J Med*. 2007;357:1316–25.
3. Meijers B, Zadora W, Lowenstein J. A historical perspective on uremia and uremic toxins. *Toxins*. 2024;16.
4. Vanholder R, De Smet R, Glorieux G, et al. Review on uremic toxins: classification, concentration, and interindividual variability. *Kidney Int*. 2003;63:1934–1943.
5. Duranton F, Cohen G, De Smet R, et al. Normal and pathologic concentrations of uremic toxins. *J Am Soc Nephrol*. 2012;23:1258–1270.
6. Rhee EP, Souza A, Farrell L, et al. Metabolite profiling identifies markers of uremia. *J Am Soc Nephrol*. 2010;21:1041–1051.
7. Kikuchi K, Itoh Y, Tateoka R, Ezawa A, Murakami K, Niwa T. Metabolomic analysis of uremic toxins by liquid chromatography/electrospray ionization-tandem mass spectrometry. *J Chromatogr B Analyt Technol Biomed Life Sci*. 2010;878:1662–1668.
8. Sirich TL, Aronov PA, Plummer NS, Hostetter TH, Meyer TW. Numerous protein-bound solutes are cleared by the kidney with high efficiency. *Kidney Int*. 2013;84:585–590.
9. Tanaka H, Sirich TL, Plummer NS, Weaver DS, Meyer TW. An enlarged profile of uremic solutes. *PLoS One*. 2015;10:e0135657.
10. Rosner MH, Reis T, Husain-Syed F, et al. Classification of uremic toxins and their role in kidney failure. *Clin J Am Soc Nephrol*. 2021;16:1918–1928.
11. J B, ed. *Uremic Toxicity*. Baltimore: Williams & Wilkins; 1997.
12. Reis T, Hutchison C, de Assis Rocha Neves F, et al. Rationale for a new classification of solutes of interest in chronic kidney disease and hemodialysis. *Blood Purif*. 2023;52:242–254.
13. Depner TA. Uremic toxicity: urea and beyond. *Semin Dial*. 2001;14:246–251.
14. Niwa T. Recent progress in the analysis of uremic toxins by mass spectrometry. *J Chromatogr B Analyt Technol Biomed Life Sci*. 2009;877:2600–2606.
15. P T, ed. *Studies in the Pathogenesis of Uremic Encephalopathy*. Stuttgart, Germany: Georg Thieme Verlag; 1972.
16. Johnson WJ, Hagge WW, Wagoner RD, Dinapoli RP, Rosevear JW. Effects of urea loading in patients with far-advanced renal failure. *Mayo Clin Proc*. 1972;47:21–29.
17. Schreiner GMJ. *Biochemistry of Uremia. Uremia*. Springfield, IL: Charles C Thomas; 1960:55–85.
18. Richards P, Brown CL. Urea metabolism in an azotaemic woman with normal renal function. *Lancet*. 1975;2:207–209.
19. Conte G, Dal Canton A, Terribile M, et al. Renal handling of urea in subjects with persistent azotemia and normal renal function. *Kidney Int*. 1987;32:721–727.
20. Hsu CH, Kurtz TW, Massari PU, Ponze SA, Chang BS. Familial azotemia. Impaired urea excretion despite normal renal function. *N Engl J Med*. 1978;298:117–121.
21. Andersen F. Final report of the safety assessment of urea. *Int J Toxicol*. 2005;24:1–56.
22. Wang Z, Nicholls SJ, Rodriguez ER, et al. Protein carbamylation links inflammation, smoking, uremia and atherogenesis. *Nat Med*. 2007;13:1176–1184.
23. Horkko S, Savolainen MJ, Kervinen K, Kesaniemi YA. Carbamylation-induced alterations in low-density lipoprotein metabolism. *Kidney Int*. 1992;41:1175–1181.
24. Shapiro RJ. Catabolism of low-density lipoprotein is altered in experimental chronic renal failure. *Metab, Clin Exp*. 1993;42:162–169.
25. Apostolov EO, Ray D, Savenka AV, Shah SV, Basnakian AG. Chronic uremia stimulates LDL carbamylation and atherosclerosis. *J Am Soc Nephrol*. 2010;21:1852–1857.
26. Horkko S, Huttunen K, Kesaniemi YA. Decreased clearance of low-density lipoprotein in uremic patients under dialysis treatment. *Kidney Int*. 1995;47:1732–1740.
27. Hörkkö S, Huttunen K, Kervinen K, Kesäniemi YA. Decreased clearance of uraemic and mildly carbamylated low-density lipoprotein. *Eur J Clin Invest*. 1994;24:105–113.
28. Apostolov EO, Shah SV, Ray D, Basnakian AG. Scavenger receptors of endothelial cells mediate the uptake and cellular proatherogenic effects of carbamylated LDL. *Arterioscler Thromb Vasc Biol*. 2009;29:1622–1630.
29. Apostolov EO, Shah SV, Ok E, Basnakian AG. Carbamylated low-density lipoprotein induces monocyte adhesion to endothelial cells through intercellular adhesion molecule-1 and vascular cell adhesion molecule-1. *Arterioscler Thromb Vasc Biol*. 2007;27:826–832.
30. Apostolov EO, Basnakian AG, Yin X, Ok E, Shah SV. Modified LDLs induce proliferation-mediated death of human vascular endothelial cells through MAPK pathway. *Am J Physiol Heart Circ Physiol*. 2007;292:H1836–H1846.
31. Ok E, Basnakian AG, Apostolov EO, Barri YM, Shah SV. Carbamylated low-density lipoprotein induces death of endothelial cells: a link to atherosclerosis in patients with kidney disease. *Kidney Int*. 2005;68:173–178.
32. Asci G, Basci A, Shah SV, et al. Carbamylated low-density lipoprotein induces proliferation and increases adhesion molecule expression of human coronary artery smooth muscle cells. *Nephrology*. 2008;13:480–486.
33. Jankowski V, Saritas T, Kjolby M, et al. Carbamylated sortilin associates with cardiovascular calcification in patients with chronic kidney disease. *Kidney Int*. 2022;101:574–584.
34. Doué M, Okwieka A, Berquand A, et al. Carbamylation of elastic fibers is a molecular substratum of aortic stiffness. *Sci Rep*. 2021;11:17827.
35. Alesutan I, Luong TTD, Schelski N, et al. Circulating uromodulin inhibits vascular calcification by interfering with pro-inflammatory cytokine signalling. *Cardiovasc Res*. 2021;117:930–941.
36. Mori D, Matsui I, Shimomura A, et al. Protein carbamylation exacerbates vascular calcification. *Kidney Int*. 2018;94:72–90.
37. Shaykh M, Pegoraro AA, Mo W, Arruda JA, Dunea G, Singh AK. Carbamylated proteins activate glomerular mesangial cells and stimulate collagen deposition. *J Lab Clin Med*. 1999;133:302–308.
38. Gross ML, Piecha G, Bierhaus A, et al. Glycated and carbamylated albumin are more “nephrotoxic” than unmodified albumin in the amphibian kidney. *Am J Physiol Ren Physiol*. 2011;301:F476–F485.
39. Jaisson S, Kazes I, Desmons A, et al. Homocitrulline as marker of protein carbamylation in hemodialyzed patients. *Clin Chim Acta*. 2016;460:5–10.
40. Awwad A, Rhee EP, Grams M, et al. Comparative CKD risk prediction using homocitrulline and carbamylated albumin: two circulating markers of protein carbamylation. *BMC Nephrol*. 2024;25:185.
41. Berg AH, Drechsler C, Wenger J, et al. Carbamylation of serum albumin as a risk factor for mortality in patients with kidney failure. *Sci Transl Med*. 2013;5:175ra29.
42. Kalim S, Trottier CA, Wenger JB, et al. Longitudinal changes in protein carbamylation and mortality risk after initiation of hemodialysis. *Clin J Am Soc Nephrol*. 2016;11:1809–1816.
43. Kalim S, Zhao S, Tang M, et al. Protein carbamylation and the risk of ESKD in patients with CKD. *J Am Soc Nephrol*. 2023;34:876–885.
44. Perl J, Kalim S, Wald R, et al. Reduction of carbamylated albumin by extended hemodialysis. *Hemodial Int*. 2016;20:510–521.
45. Eknoyan G, Beck GJ, Cheung AK, et al. Effect of dialysis dose and membrane flux in maintenance hemodialysis. *N Engl J Med*. 2002;347:2010–2019.
46. Nagata Y, Akino T, Ohno K, et al. Free D-amino acids in human plasma in relation to senescence and renal diseases. *Clin Sci*. 1987;73:105–108.
47. Bruckner H, Hausch M. Gas chromatographic characterization of free D-amino acids in the blood serum of patients with renal disorders and of healthy volunteers. *J Chromatogr*. 1993;614:7–17.
48. Friedman M. Chemistry, nutrition, and microbiology of D-amino acids. *J Agric Food Chem*. 1999;47:3457–3479.
49. Silbernagl S, Volker K, Dantzler WH. D-Serine is reabsorbed in rat renal pars recta. *Am J Physiol*. 1999;276:F857–F863.
50. Hasegawa H, Matsukawa T, Shinohara Y, Konno R, Hashimoto T. Role of renal D-amino-acid oxidase in pharmacokinetics of D-leucine. *Am J Physiol Endocrinol Metab*. 2004;287:E160–E165.
51. Nagata Y, Masui R, Akino T. The presence of free D-serine, D-alanine and D-proline in human plasma. *Experientia*. 1992;48:986–988.
52. Young GA, Kendall S, Brownjohn AM. D-Amino acids in chronic renal failure and the effects of dialysis and urinary losses. *Amino Acids*. 1994;6:283–293.
53. Fuchs SA, Berger R, Klomp LW, de Koning TJ. D-amino acids in the central nervous system in health and disease. *Mol Genet Metab*. 2005;85:168–180.

54. Nagata Y, Konno R, Niwa A. Amino acid levels in D-alanine-administered mutant mice lacking D-amino acid oxidase. *Metab, Clin Exp*. 1994;43:1153–1157.
55. Hashimoto A, Nishikawa T, Konno R, et al. Free D-serine, D-aspartate and D-alanine in central nervous system and serum in mutant mice lacking D-amino acid oxidase. *Neurosci Lett*. 1993;152:33–36.
56. Harper AE, Benevenga NJ, Wohlhueter RM. Effects of ingestion of disproportionate amounts of amino acids. *Physiol Rev*. 1970;50:428–558.
57. Adibi SA. Renal assimilation of oligopeptides: physiological mechanisms and metabolic importance. *Am J Physiol*. 1997;272:E723–E736.
58. Daniel H, Herget M. Cellular and molecular mechanisms of renal peptide transport. *Am J Physiol*. 1997;273:F1–F8.
59. Druml W, Lochs H, Roth E, Hubl W, Balcke P, Lenz K. Utilization of tyrosine dipeptides and acetyltyrosine in normal and uremic humans. *Am J Physiol*. 1991;260:E280–E285.
60. Verroust PJ, Birn H, Nielsen R, Kozyraki R, Christensen EI. The tandem endocytic receptors megalin and cubilin are important proteins in renal pathology. *Kidney Int*. 2002;62:745–756.
61. Maack T, Johnson V, Kau ST, Figueiredo J, Sigulem D. Renal filtration, transport, and metabolism of low-molecular-weight proteins: a review. *Kidney Int*. 1979;16:251–270.
62. Inker LA, Wyatt C, Creamer R, et al. Performance of creatinine and cystatin C GFR estimating equations in an HIV-positive population on antiretrovirals. *J Acquir Immune Defic Syndr*. 2012;61:302–309.
63. Norden AG, Sharratt P, Cutillas PR, Cramer R, Gardner SC, Unwin RJ. Quantitative amino acid and proteomic analysis: very low excretion of polypeptides >750 Da in normal urine. *Kidney Int*. 2004;66:1994–2003.
64. Canaud B, Morena M, Cristol JP, Krieter D. Beta2-microglobulin, a uremic toxin with a double meaning. *Kidney Int*. 2006;69:1297–1299.
65. Kabanda A, Jadoul M, Pochet JM, Lauwerys R, van Ypersele de Strihou C, Bernard A. Determinants of the serum concentrations of low molecular weight proteins in patients on maintenance hemodialysis. *Kidney Int*. 1994;45:1689–1696.
66. Penders J, Delanghe JR. Alpha 1-microglobulin: clinical laboratory aspects and applications. *Clin Chim Acta*. 2004;346:107–118.
67. Jaconi S, Saurat JH, Siegenthaler G. Analysis of normal and truncated holo- and apo-retinol-binding protein (RBP) in human serum: altered ratios in chronic renal failure. *Eur J Endocrinol*. 1996;134:576–582.
68. White CA, Akbari A, Doucette S, et al. Estimating GFR using serum beta trace protein: accuracy and validation in kidney transplant and pediatric populations. *Kidney Int*. 2009;76:784–791.
69. Hermans C, Dong P, Robin M, et al. Determinants of serum levels of surfactant proteins A and B and Clara cell protein CC16. *Biomarkers*. 2003;8:461–471.
70. Ziegler MG, Kennedy B, Morrissey E, O'Connor DT. Norepinephrine clearance, chromogranin A and dopamine beta hydroxylase in renal failure. *Kidney Int*. 1990;37:1357–1362.
71. Cohen G, Horl WH. Free immunoglobulin light chains as a risk factor in renal and extrarenal complications. *Semin Dial*. 2009;22:369–372.
72. Sjostrom P, Tidman M, Jones I. Determination of the production rate and non-renal clearance of cystatin C and estimation of the glomerular filtration rate from the serum concentration of cystatin C in humans. *Scand J Clin Lab Invest*. 2005;65:111–124.
73. Birn H. The kidney in vitamin B12 and folate homeostasis: characterization of receptors for tubular uptake of vitamins and carrier proteins. *Am J Physiol Ren Physiol*. 2006;291:F22–F36.
74. Yoshimoto A, Mori K, Sugawara A, et al. Plasma ghrelin and desacyl ghrelin concentrations in renal failure. *J Am Soc Nephrol*. 2002;13:2748–2752.
75. Axelsson J, Bergsten A, Qureshi AR, et al. Elevated resistin levels in chronic kidney disease are associated with decreased glomerular filtration rate and inflammation, but not with insulin resistance. *Kidney Int*. 2006;69:596–604.
76. Don BR, Rosales LM, Levine NW, Mitch W, Kaysen GA. Leptin is a negative acute phase protein in chronic hemodialysis patients. *Kidney Int*. 2001;59:1114–1120.
77. Lu XL, Xiao ZH, Yang MY, Zhu YM. Diagnostic value of serum procalcitonin in patients with chronic renal insufficiency: a systematic review and meta-analysis. *Nephrol Dial Transplant*. 2013;28:122–129.
78. Colbert G, Jain N, de Lemos JA, Hedayati SS. Utility of traditional circulating and imaging-based cardiac biomarkers in patients with predialysis CKD. *Clin J Am Soc Nephrol*. 2015;10:515–529.
79. Richter R, Schulz-Knappe P, Schrader M, et al. Composition of the peptide fraction in human blood plasma: database of circulating human peptides. *J Chromatogr B Biomed Sci Appl*. 1999;726:25–35.
80. Li G, Chu J, Liu X, Yuan Z. Separation, identification of uremic middle molecules, and preliminary study on their toxicity. *Clin Chim Acta*. 2004;350:89–98.
81. Kaplan B, Cojocaru M, Unsworth E, Knecht A, Martin BM. Search for peptidic "middle molecules" in uremic sera: isolation and chemical identification of fibrinogen fragments. *J Chromatogr B Analyt Technol Biomed Life Sci*. 2003;796:141–153.
82. Naseeb U, Shafqat J, Jagerbrink T, et al. Proteome patterns in uremic plasma. *Blood Purif*. 2008;26:561–568.
83. Weissinger EM, Kaiser T, Meert N, et al. Proteomics: a novel tool to unravel the patho-physiology of uraemia. *Nephrol Dial Transplant*. 2004;19:3068–3077.
84. Deppisch RM, Beck W, Goehl H, Ritz E. Complement components as uremic toxins and their potential role as mediators of microinflammation. *Kidney Int Suppl*. 2001;78:S271–S277.
85. Jadoul M, Drueke TB. beta2 microglobulin amyloidosis: an update 30 years later. *Nephrol Dial Transplant*. 2016;31:507–509.
86. Eloot S, Torremans A, De Smet R, et al. Kinetic behavior of urea is different from that of other water-soluble compounds: the case of the guanidino compounds. *Kidney Int*. 2005;67:1566–1575.
87. Marescau B, Nagels G, Possemiers I, et al. Guanidino compounds in serum and urine of nondialyzed patients with chronic renal insufficiency. *Metab, Clin Exp*. 1997;46:1024–1031.
88. De Deyn PP, Vanholder R, Eloot S, Glorieux G. Guanidino compounds as uremic (neuro)toxins. *Semin Dial*. 2009;22:340–345.
89. Wyss M, Kaddurah-Daouk R. Creatine and creatinine metabolism. *Physiol Rev*. 2000;80:1107–1213.
90. Nakamura K, Ienaga K, Nakano K, et al. Creatol, a creatinine metabolite, as a useful determinant of renal function. *Nephron*. 1994;66:140–146.
91. Yokozawa T, Fujitsuka N, Oura H. Studies on the precursor of methylguanidine in rats with renal failure. *Nephron*. 1991;58:90–94.
92. Gonella M, Barsotti G, Lupetti S, Giovannetti S. Factors affecting the metabolic production of methylguanidine. *Clin Sci Mol Med*. 1975;48:341–347.
93. Cohen BD. Methyl group deficiency and guanidino production in uremia. *Mol Cell Biochem*. 2003;244:31–36.
94. Aoyagi K, Shahrzad S, Iida S, et al. Role of nitric oxide in the synthesis of guanidinosuccinic acid, an activator of the N-methyl-D-aspartate receptor. *Kidney Int Suppl*. 2001;78:S93–S96.
95. Kopple JD, Gordon SI, Wang M, Swendseid ME. Factors affecting serum and urinary guanidinosuccinic acid levels in normal and uremic subjects. *J Lab Clin Med*. 1977;90:303–311.
96. Noris M, Remuzzi G. Uremic bleeding: closing the circle after 30 years of controversies? *Blood*. 1999;94:2569–2574.
97. Barsotti G, Bevilacqua G, Morelli E, Cappelli P, Balestri PL, Giovannetti S. Toxicity arising from guanidine compounds: role of methylguanidine as a uremic toxin. *Kidney Int Suppl*. 1975:299–301.
98. Glorieux GL, Dhondt AW, Jacobs P, et al. In vitro study of the potential role of guanidines in leukocyte functions related to atherogenesis and infection. *Kidney Int*. 2004;65:2184–2192.
99. De Deyn PP, D'Hooge R, Van Bogaert PP, Marescau B. Endogenous guanidino compounds as uremic neurotoxins. *Kidney Int Suppl*. 2001;78:S77–S83.
100. Siroen MP, van der Sijp JR, Teerlink T, van Schaik C, Nijveldt RJ, van Leeuwen PA. The human liver clears both asymmetric and symmetric dimethylarginine. *Hepatology*. 2005;41:559–565.
101. Schepers E, Speer T, Bode-Boger SM, Fliser D, Kielstein JT. Dimethylarginines ADMA and SDMA: the real water-soluble small toxins? *Semin Nephrol*. 2014;34:97–105.
102. Ronden RA, Houben AJ, Teerlink T, et al. Reduced renal plasma clearance does not explain increased plasma asymmetric dimethylarginine in hypertensive subjects with mild to moderate renal insufficiency. *Am J Physiol Ren Physiol*. 2012;303:F149–F156.
103. Kielstein JT, Boger RH, Bode-Boger SM, et al. Marked increase of asymmetric dimethylarginine in patients with incipient primary chronic renal disease. *J Am Soc Nephrol*. 2002;13:170–176.
104. Powell GM, Miller JJ, Olavesen AH, Curtis CG. Liver as major organ of phenol detoxication? *Nature*. 1974;252:234–235.

105. Lock EA, Reed CJ. Xenobiotic metabolizing enzymes of the kidney. *Toxicol Pathol*. 1998;26:18–25.
106. Seligson D, Bluemle L, Webster G, Senesky D. Organic acids in body fluids of the uremic patient. *J Clin Investig*. 1959:1042–1043.
107. Wengle B, Hellstrom K. Volatile phenols in serum of uraemic patients. *Clin Sci*. 1972;43:493–498.
108. Niwa T, Maeda K, Ohki T, et al. Profiling of uremic ultrafiltrate using high resolution gas chromatography-mass spectrometry - identification of 6 polyphenols. *Clin Chim Acta*. 1980;108:113–119.
109. Niwa T, Maeda K, Ohki T, Saito A, Kobayashi K. A gas chromatographic-mass spectrometric analysis for phenols in uremic serum. *Clin Chim Acta*. 1981;110:51–57.
110. Schoots AC, De Vries PM, Thiemann R, Hazejager WA, Visser SL, Oe PL. Biochemical and neurophysiological parameters in hemodialyzed patients with chronic renal failure. *Clin Chim Acta*. 1989;185:91–107.
111. Wardle EN. How toxic are phenols? *Kidney Int Suppl*. 1978:S13–S15.
112. Bammens B, Evenepoel P, Keuleers H, Verbeke K, Vanrenterghem Y. Free serum concentrations of the protein-bound retention solute p-cresol predict mortality in hemodialysis patients. *Kidney Int*. 2006;69:1081–1087.
113. Armstrong MD, Chao FC, Parker VJ, Wall PE. Endogenous formation of hippuric acid. *Proc Soc Exp Biol Med*. 1955;90:675–679.
114. Mulder TP, Rietveld AG, van Amelsvoort JM. Consumption of both black tea and green tea results in an increase in the excretion of hippuric acid into urine. *Am J Clin Nutr*. 2005;81:256S–60S.
115. Remer T, Manz F. Paleolithic diet, sweet potato eaters, and potential renal acid load. *Am J Clin Nutr*. 2003;78:802–803; author reply 3-4.
116. Cathcart-Rake W, Porter R, Whittier F, Stein P, Carey M, Grantham J. Effect of diet on serum accumulation and renal excretion of aryl acids and secretory activity in normal and uremic man. *Am J Clin Nutr*. 1975;28:1110–1115.
117. Mitch WE, Brusilow S. Benzoate-induced changes in glycine and urea metabolism in patients with chronic renal failure. *J Pharmacol Exp Ther*. 1982;222:572–575.
118. Meijers BK, Evenepoel P. The gut-kidney axis: indoxyl sulfate, p-cresyl sulfate and CKD progression. *Nephrol Dial Transplant*. 2011;26:759–761.
119. Meyer TW, Hostetter TH. Uremic solutes from colon microbes. *Kidney Int*. 2012;81:949–954.
120. de Loor H, Bammens B, Evenepoel P, De Preter V, Verbeke K. Gas chromatographic-mass spectrometric analysis for measurement of p-cresol and its conjugated metabolites in uremic and normal serum. *Clin Chem*. 2005;51:1535–1538.
121. Martinez AW, Recht NS, Hostetter TH, Meyer TW. Removal of P-cresol sulfate by hemodialysis. *J Am Soc Nephrol*. 2005;16:3430–3436.
122. Bammens B, Evenepoel P, Verbeke K, Vanrenterghem Y. Time profiles of peritoneal and renal clearances of different uremic solutes in incident peritoneal dialysis patients. *Am J Kidney Dis*. 2005;46:512–519.
123. Lesaffer G, De Smet R, Lameire N, Dhondt A, Duym P, Vanholder R. Intradialytic removal of protein-bound uraemic toxins: role of solute characteristics and of dialyser membrane. *Nephrol Dial Transplant*. 2000;15:50–57.
124. Sirich TL, Luo FJ, Plummer NS, Hostetter TH, Meyer TW. Selectively increasing the clearance of protein-bound uremic solutes. *Nephrol Dial Transplant*. 2012;27:1574–1579.
125. Meijers BK, Bammens B, De Moor B, Verbeke K, Vanrenterghem Y, Evenepoel P. Free p-cresol is associated with cardiovascular disease in hemodialysis patients. *Kidney Int*. 2008;73:1174–1180.
126. Meijers BK, Verbeke K, Dehaen W, Vanrenterghem Y, Hoylaerts MF, Evenepoel P. The uremic retention solute p-cresyl sulfate and markers of endothelial damage. *Am J Kidney Dis*. 2009;54:891–901.
127. Vanholder R, Schepers E, Pletinck A, Nagler EV, Glorieux G. The uremic toxicity of indoxyl sulfate and p-cresyl sulfate: a systematic review. *J Am Soc Nephrol*. 2014;25:1897–1907.
128. Schepers E, Meert N, Glorieux G, Goeman J, Van der Eycken J, Vanholder R. P-cresylsulphate, the main in vivo metabolite of p-cresol, activates leucocyte free radical production. *Nephrol Dial Transplant*. 2007;22:592–596.
129. Jones MR, Kopple JD, Swendseid ME. Phenylalanine metabolism in uremic and normal man. *Kidney Int*. 1978;14:169–179.
130. Record NB, Gallagher BB, Glaser GH. Phenolic acids in experimental uremia. II. Relationship of phenolic acid structure to seizure threshold in uremia. *Arch Neurol*. 1969;21:395–400.
131. Zimmerman L, Jornvall H, Bergstrom J. Phenylacetylglutamine and hippuric acid in uremic and healthy subjects. *Nephron*. 1990;55:265–271.
132. Poesen R, Claes K, Evenepoel P, et al. Microbiota-derived phenylacetylglutamine associates with overall mortality and cardiovascular disease in patients with CKD. *J Am Soc Nephrol*. 2016;27:3479–3487.
133. Kurella Tamura M, Chertow GM, Depner TA, et al. Metabolic profiling of impaired cognitive function in patients receiving dialysis. *J Am Soc Nephrol*. 2016;27:3780–3787.
134. van Haard PM. Chromatography of urinary indole derivatives. *J Chromatogr*. 1988;429:59–94.
135. Dou L, Bertrand E, Cerini C, et al. The uremic solutes p-cresol and indoxyl sulfate inhibit endothelial proliferation and wound repair. *Kidney Int*. 2004;65:442–451.
136. Dou L, Jourde-Chiche N, Faure V, et al. The uremic solute indoxyl sulfate induces oxidative stress in endothelial cells. *J Thromb Haemostasis*. 2007;5:1302–1308.
137. Nii-Kono T, Iwasaki Y, Uchida M, et al. Indoxyl sulfate induces skeletal resistance to parathyroid hormone in cultured osteoblastic cells. *Kidney Int*. 2007;71:738–743.
138. Arinze NV, Yin W, Lotfollahzadeh S, et al. Tryptophan metabolites suppress the Wnt pathway and promote adverse limb events in chronic kidney disease. *J Clin Investig*. 2022;132:e142260.
139. Miyazaki T, Aoyama I, Ise M, Seo H, Niwa T. An oral sorbent reduces overload of indoxyl sulphate and gene expression of TGF-beta1 in uraemic rat kidneys. *Nephrol Dial Transplant*. 2000;15:1773–1781.
140. Schulman G, Berl T, Beck GJ, et al. Randomized placebo-controlled EPPIC trials of AST-120 in CKD. *J Am Soc Nephrol*. 2015;26:1732–1746.
141. Saito A, Niwa T, Maeda K, Kobayashi K, Yamamoto Y, Ohta K. Tryptophan and indolic tryptophan metabolites in chronic renal failure. *Am J Clin Nutr*. 1980;33:1402–1406.
142. Ludwig GD, Senesky D, Bluemle Jr LW, Elkinton JR. Indoles in ureamia: identification by countercurrent distribution and paper chromatography. *Am J Clin Nutr*. 1968;21:436–450.
143. Shivanna S, Kolandaivelu K, Shashar M, et al. The aryl hydrocarbon receptor is a critical regulator of tissue factor stability and an antithrombotic target in uremia. *J Am Soc Nephrol*. 2016;27:189–201.
144. Gondouin B, Cerini C, Dou L, et al. Indolic uremic solutes increase tissue factor production in endothelial cells by the aryl hydrocarbon receptor pathway. *Kidney Int*. 2013;84:733–744.
145. Saito K, Fujigaki S, Heyes MP, et al. Mechanism of increases in L-kynurenine and quinolinic acid in renal insufficiency. *Am J Physiol Ren Physiol*. 2000;279:F565–F572.
146. Pawlak D, Pawlak K, Malyszko J, Mysliwiec M, Buczko W. Accumulation of toxic products degradation of kynurenine in hemodialyzed patients. *Int Urol Nephrol*. 2001;33:399–404.
147. Shim CK, Sawada Y, Iga T, Hanano M. Estimation of renal secretory function for organic cations by endogenous N1-methylnicotinamide in rats with experimental renal failure. *J Pharmacokinet Biopharm*. 1984;12:23–42.
148. Ponda MP, Quan Z, Melamed ML, Raff A, Meyer TW, Hostetter TH. Methylamine clearance by haemodialysis is low. *Nephrol Dial Transplant*. 2010;25:1608–1613.
149. Quan Z, Ponda M, Melamed M, et al. Derivatization of methylamine and ethylamine followed by LC and fluorescence detection for measurement of urinary clearance. *Chromatographia*. 2010;71:411–418.
150. Smith JL, Wishnok JS, Deen WM. Metabolism and excretion of methylamines in rats. *Toxicol Appl Pharmacol*. 1994;125:296–308.
151. Ihle BU, Cox RW, Dunn SR, Simenhoff ML. Determination of body burden of uremic toxins. *Clin Nephrol*. 1984;22:82–89.
152. Wright SH, Dantzler WH. Molecular and cellular physiology of renal organic cation and anion transport. *Physiol Rev*. 2004;84:987–1049.
153. Weiner ID, Verlander JW. Role of NH3 and NH4+ transporters in renal acid-base transport. *Am J Physiol Ren Physiol*. 2011;300:F11–F23.

154. Roch–Ramel F, Besseghir K, Murer H. Renal excretion and tubular transport of organic anions and cations. *Compr Physiol*. 2010:2189–2262.
155. Aronov PA, Luo FJ, Plummer NS, et al. Colonic contribution to uremic solutes. *J Am Soc Nephrol*. 2011;22:1769–1776.
156. Zeisel SH, Wishnok JS, Blusztajn JK. Formation of methylamines from ingested choline and lecithin. *J Pharmacol Exp Ther*. 1983;225:320–324.
157. Zhang AQ, Mitchell SC, Ayesh R, Smith RL. Dimethylamine formation in man. *Biochem Pharmacol*. 1993;45:2185–2188.
158. Zhang AQ, Mitchell SC, Smith RL. Dietary precursors of trimethylamine in man: a pilot study. *Food Chem Toxicol*. 1999;37:515–520.
159. Simenhoff ML, Saukkonen JJ, Burke JF, Wesson Jr LG, Schaedler RW, Gordon SJ. Bacterial populations of the small intestine in uremia. *Nephron*. 1978;22:63–68.
160. Mitchell SC, Zhang AQ. Methylamine in human urine. *Clin Chim Acta*. 2001;312:107–114.
161. Pirisino R, Ghelardini C, Pacini A, Galeotti N, Raimondi L. Methylamine, but not ammonia, is hypophagic in mouse by interaction with brain Kv1.6 channel subtype. *Br J Pharmacol*. 2004;142:381–389.
162. Melamed ML, Plantinga L, Shafi T, et al. Retained organic solutes, patient characteristics and all-cause and cardiovascular mortality in hemodialysis: results from the retained organic solutes and clinical outcomes (ROSCO) investigators. *BMC Nephrol*. 2013;14:134.
163. Simenhoff ML, Burke JF, Saukkonen JJ, Ordinario AT, Doty R. Biochemical profile or uremic breath. *N Engl J Med*. 1977;297:132–135.
164. Raff AC, Lieu S, Melamed ML, et al. Relationship of impaired olfactory function in ESRD to malnutrition and retained uremic molecules. *Am J Kidney Dis*. 2008;52:102–110.
165. Tang WH, Wang Z, Levison BS, et al. Intestinal microbial metabolism of phosphatidylcholine and cardiovascular risk. *N Engl J Med*. 2013;368:1575–1584.
166. Hai X, Landeras V, Dobre MA, DeOreo P, Meyer TW, Hostetter TH. Mechanism of prominent trimethylamine oxide (TMAO) accumulation in hemodialysis patients. *PLoS One*. 2015;10:e0143731.
167. Bain MA, Faull R, Fornasini G, Milne RW, Evans AM. Accumulation of trimethylamine and trimethylamine-N-oxide in end-stage renal disease patients undergoing haemodialysis. *Nephrol Dial Transplant*. 2006;21:1300–1304.
168. Stubbs JR, House JA, Ocque AJ, et al. Serum trimethylamine-N-oxide is elevated in CKD and correlates with coronary atherosclerosis burden. *J Am Soc Nephrol*. 2016;27:305–313.
169. Shafi T, Powe NR, Meyer TW, et al. Trimethylamine N-oxide and cardiovascular events in hemodialysis patients. *J Am Soc Nephrol*. 2017;28:321–331.
170. Niwa T, Yamamoto N, Maeda K, Yamada K, Ohki T, Mori M. Gas chromatographic–mass spectrometric analysis of polyols in urine and serum of uremic patients. Identification of new deoxyalditols and inositol isomers. *J Chromatogr*. 1983;277:25–39.
171. Niwa T, Tohyama K, Kato Y. Analysis of polyols in uremic serum by liquid chromatography combined with atmospheric pressure chemical ionization mass spectrometry. *J Chromatogr*. 1993;613:9–14.
172. Niwa T, Sobue G, Maeda K, Mitsuma T. Myoinositol inhibits proliferation of cultured Schwann cells: evidence for neurotoxicity of myoinositol. *Nephrol Dial Transplant*. 1989;4:662–666.
173. Beberashvili I, Erlich A, Azar A, et al. Longitudinal study of serum uric acid, nutritional status, and mortality in maintenance hemodialysis patients. *Clin J Am Soc Nephrol*. 2016;11:1015–1023.
174. Niwa T, Takeda N, Yoshizumi H. RNA metabolism in uremic patients: accumulation of modified ribonucleosides in uremic serum. Technical note. *Kidney Int*. 1998;53:1801–1806.
175. Daniewska-Michalska D, Motyl T, Gellert R, et al. Efficiency of hemodialysis of pyrimidine compounds in patients with chronic renal failure. *Nephron*. 1993;64:193–197.
176. Mydlik M, Derzsiova K. Renal replacement therapy and secondary hyperoxalemia in chronic renal failure. *Kidney Int Suppl*. 2001;78:S304–S307.
177. Holmes RP, Goodman HO, Assimos DG. Contribution of dietary oxalate to urinary oxalate excretion. *Kidney Int*. 2001;59:270–276.
178. Fehrman-Ekholm I, Lotsander A, Logan K, Dunge D, Odar-Cederlof I, Kallner A. Concentrations of vitamin C, vitamin B12 and folic acid in patients treated with hemodialysis and on-line hemodiafiltration or hemofiltration. *Scand J Urol Nephrol*. 2008;42:74–80.
179. Canavese C, Marangella M, Stratta P. Think of oxalate when using ascorbate supplementation to optimize iron therapy in dialysis patients. *Nephrol Dial Transplant*. 2008;23:1463–1464.
180. Yokoyama K, Tajima M, Yoshida H, et al. Plasma pteridine concentrations in patients with chronic renal failure. *Nephrol Dial Transplant*. 2002;17:1032–1036.
181. Williams RH, Shaykh M, Ahmed S, et al. Purification and biochemical characterization of xanthopterin from patients with chronic renal failure. I. Isolation, purification and preliminary characterization. *Clin Biochem*. 1991;24:399–406.
182. Fanti P, Stephenson TJ, Kaariainen IM, et al. Serum isoflavones and soya food intake in Japanese, Thai and American end-stage renal disease patients on chronic haemodialysis. *Nephrol Dial Transplant*. 2003;18:1862–1868.
183. Huang Y, Sun H, Frassetto L, Benet LZ, Lin ET. Liquid chromatographic tandem mass spectrometric assay for the uremic toxin 3-carboxy-4-methyl-5-propyl-2-furanpropionic acid in human plasma. *Rapid Commun Mass Spectrom*. 2006;20:1611–1614.
184. Kalim S, Smoyer WE. The promise and challenges of metabolomic studies in pediatric CKD. *Clin J Am Soc Nephrol*. 2024;19:823–825. CJASN.
185. Wulczyn KE, Shafi T, Anderson A, et al. Metabolites associated with uremic symptoms in patients with CKD: findings from the chronic renal insufficiency cohort (CRIC) study. *Am J Kidney Dis*. 2024;84:49–61.
186. Al AS, Myint L, Guallar E, et al. A metabolomics approach to identify metabolites associated with mortality in patients receiving maintenance hemodialysis. *Kidney Int Rep*. 2024;9:2718–2726.
187. Kalim S, Clish CB, Wenger J, et al. A plasma long-chain acylcarnitine predicts cardiovascular mortality in incident dialysis patients. *J Am Heart Assoc*. 2013;2:e000542.
188. Moorthi RN, Moe SM, O'Connell T, et al. Plasma metabolites and physical function in patients undergoing hemodialysis. *Sci Rep*. 2024;14:8427.
189. Wen D, Zheng Z, Surapaneni A, et al. Metabolite profiling of CKD progression in the chronic renal insufficiency cohort study. *JCI Insight*. 2022;7:e161696.
190. Kalim S, Wald R, Yan AT, et al. Extended duration nocturnal hemodialysis and changes in plasma metabolite profiles. *Clin J Am Soc Nephrol*. 2018;13:436–444.
191. Sirich TL, Fong K, Larive B, et al. Limited reduction in uremic solute concentrations with increased dialysis frequency and time in the Frequent Hemodialysis Network Daily Trial. *Kidney Int*. 2017;91:1186–1192.
192. Meyer TW, Sirich TL, Fong KD, et al. Kt/vurea and nonurea small solute levels in the hemodialysis study. *J Am Soc Nephrol*. 2016;27:3469–3478.
193. Babb AL, Ahmad S, Bergstrom J, Scribner BH. The middle molecule hypothesis in perspective. *Am J Kidney Dis*. 1981;1:46–50.
194. Babb AL, Farrell PC, Uvelli DA, Scribner BH. Hemodialyzer evaluation by examination of solute molecular spectra. *Trans Am Soc Artif Intern Organs*. 1972;18:98–105, 22.
195. Henderson LW, Colton CK, Ford CA, Bosch JP. Kinetics of hemodiafiltration. II. Clinical characterization of a new blood cleansing modality. 1975. *J Am Soc Nephrol*. 1997;8:494–508.
196. Ward RA, Greene T, Hartmann B, Samtleben W. Resistance to intercompartmental mass transfer limits beta2-microglobulin removal by post-dilution hemodiafiltration. *Kidney Int*. 2006;69:1431–1437.
197. Sirich TL. Obstacles to reducing plasma levels of uremic solutes by hemodialysis. *Semin Dial*. 2017;30:403–408.
198. Nistor I, Palmer SC, Craig JC, et al. Haemodiafiltration, haemofiltration and haemodialysis for end-stage kidney disease. *Cochrane Database Syst Rev*. 2015;2015:Cd006258.
199. Blankestijn PJ, Vernooij RWM, Hockham C, et al. Effect of hemodiafiltration or hemodialysis on mortality in kidney failure. *N Engl J Med*. 2023;389:700–709.
200. Weiner DE, Falzon L, Skoufos L, et al. Efficacy and safety of expanded hemodialysis with the theranova 400 dialyzer: a randomized controlled trial. *Clin J Am Soc Nephrol*. 2020;15:1310–1319.
201. Alarcon JC, Bunch A, Ardila F, et al. Impact of medium cut-off dialyzers on patient-reported outcomes: COREXH registry. *Blood Purif*. 2021;50:110–118.

202. Sirich TL, Meyer TW, Gondouin B, Brunet P, Niwa T. Protein-bound molecules: a large family with a bad character. *Semin Nephrol.* 2014;34:106–117.
203. Vanholder R, Schepers E, Pletinck A, Neirynck N, Glorieux G. An update on protein-bound uremic retention solutes. *J Ren Nutr.* 2012;22:90–94.
204. Bammens B, Evenepoel P, Verbeke K, Vanrenterghem Y. Removal of the protein-bound solute p-cresol by convective transport: a randomized crossover study. *Am J Kidney Dis.* 2004;44:278–285.
205. Luo FJ, Patel KP, Marquez IO, Plummer NS, Hostetter TH, Meyer TW. Effect of increasing dialyzer mass transfer area coefficient and dialysate flow on clearance of protein-bound solutes: a pilot crossover trial. *Am J Kidney Dis.* 2009;53:1042–1049.
206. Vanholder RC, Eloot S, Glorieux GL. Future avenues to decrease uremic toxin concentration. *Am J Kidney Dis.* 2016;67:664–676.
207. Maheshwari V, Tao X, Thijssen S, Kotanko P. Removal of protein-bound uremic toxins using binding competitors in hemodialysis: a narrative review. *Toxins.* 2021;13.
208. Bammens B, Evenepoel P, Verbeke K, Vanrenterghem Y. Removal of middle molecules and protein-bound solutes by peritoneal dialysis and relation with uremic symptoms. *Kidney Int.* 2003;64:2238–2243.
209. Pham NM, Recht NS, Hostetter TH, Meyer TW. Removal of the protein-bound solutes indican and p-cresol sulfate by peritoneal dialysis. *Clin J Am Soc Nephrol.* 2008;3:85–90.
210. Sirich TL, Fong K, Larive B, et al. Limited reduction in uremic solute concentrations with increased dialysis frequency and time in the Frequent Hemodialysis Network Daily Trial. *Kidney Int.* 2017;91:1186–1192.
211. Schneditz D, Daugirdas JT. Compartment effects in hemodialysis. *Semin Dial.* 2001;14:271–277.
212. Eloot S, Torremans A, De Smet R, et al. Complex compartmental behavior of small water-soluble uremic retention solutes: evaluation by direct measurements in plasma and erythrocytes. *Am J Kidney Dis.* 2007;50:279–288.
213. Kopple JD, Greene T, Chumlea WC, et al. Relationship between nutritional status and the glomerular filtration rate: results from the MDRD study. *Kidney Int.* 2000;57:1688–1703.
214. Giovannetti S, Maggiore Q. A low-nitrogen diet with proteins of high biological value for severe chronic uraemia. *Lancet.* 1964;1:1000–1003.
215. Uribarri J. The obsession with high dietary protein intake in ESRD patients on dialysis: is it justified? *Nephron.* 2000;86:105–108.
216. Ikizler TA, Burrowes JD, Byham-Gray LD, et al. KDOQI clinical practice guideline for nutrition in CKD: 2020 update. *Am J Kidney Dis.* 2020;76:S1–S107.
217. Cohen BD. Guanidinosuccinic acid in uremia. *Arch Intern Med.* 1970;126:847–850.
218. Ando A, Orita Y, Nakata K, et al. Effect of low protein diet and surplus of essential amino acids on the serum concentration and the urinary excretion of methylguanidine and guanidinosuccinic acid in chronic renal failure. *Nephron.* 1979;24:161–169.
219. Niwa T, Ise M. Indoxyl sulfate, a circulating uremic toxin, stimulates the progression of glomerular sclerosis. *J Lab Clin Med.* 1994;124:96–104.
220. Bammens B, Verbeke K, Vanrenterghem Y, Evenepoel P. Evidence for impaired assimilation of protein in chronic renal failure. *Kidney Int.* 2003;64:2196–2203.
221. Vaziri ND, Wong J, Pahl M, et al. Chronic kidney disease alters intestinal microbial flora. *Kidney Int.* 2013;83:308–315.
222. Niwa T, Emoto Y, Maeda K, Uehara Y, Yamada N, Shibata M. Oral sorbent suppresses accumulation of albumin-bound indoxyl sulphate in serum of haemodialysis patients. *Nephrol Dial Transplant.* 1991;6:105–109.
223. Evenepoel P, Meijers BK. Dietary fiber and protein: nutritional therapy in chronic kidney disease and beyond. *Kidney Int.* 2012;81:227–229.
224. Sirich TL, Plummer NS, Gardner CD, Hostetter TH, Meyer TW. Effect of increasing dietary fiber on plasma levels of colon-derived solutes in hemodialysis patients. *Clin J Am Soc Nephrol.* 2014;9:1603–1610.
225. Laville SM, Massy ZA, Kamel S, Chillon JM, Choukroun G, Liabeuf S. Intestinal chelators, sorbants, and gut-derived uremic toxins. *Toxins.* 2021;13.
226. Sirich TL, Meyer TW. Manipulating the microbiome. *Kidney Int.* 2017;91:274–276.
227. Di Iorio BR, Marzocco S, Bellasi A, et al. Nutritional therapy reduces protein carbamylation through urea lowering in chronic kidney disease. *Nephrol Dial Transplant.* 2017:gfx203–gfx.
228. Jonker JW, Wagenaar E, Van Eijl S, Schinkel AH. Deficiency in the organic cation transporters 1 and 2 (Oct1/Oct2 [Slc22a1/Slc22a2]) in mice abolishes renal secretion of organic cations. *Mol Cell Biol.* 2003;23:7902–7908.
229. Eraly SA, Vallon V, Vaughn DA, et al. Decreased renal organic anion secretion and plasma accumulation of endogenous organic anions in OAT1 knock-out mice. *J Biol Chem.* 2006;281:5072–5083.
230. Nolin TD. Altered nonrenal drug clearance in ESRD. *Curr Opin Nephrol Hypertens.* 2008;17:555–559.
231. Hosoya K, Tachikawa M. Roles of organic anion/cation transporters at the blood-brain and blood-cerebrospinal fluid barriers involving uremic toxins. *Clin Exp Nephrol.* 2011;15:478–485.
232. Ladda MA, Goralski KB. The effects of CKD on cytochrome P450-mediated drug metabolism. *Adv Chron Kidney Dis.* 2016;23:67–75.
233. Himmelfarb J. Uremic toxicity, oxidative stress, and hemodialysis as renal replacement therapy. *Semin Dial.* 2009;22:636–643.
234. Yoon JW, Pahl MV, Vaziri ND. Spontaneous leukocyte activation and oxygen-free radical generation in end-stage renal disease. *Kidney Int.* 2007;71:167–172.
235. Agalou S, Ahmed N, Babaei-Jadidi R, Dawnay A, Thornalley PJ. Profound mishandling of protein glycation degradation products in uremia and dialysis. *J Am Soc Nephrol.* 2005;16:1471–1485.
236. Capeillere-Blandin C, Gausson V, Descamps-Latscha B, Witko-Sarsat V. Biochemical and spectrophotometric significance of advanced oxidized protein products. *Biochim Biophys Acta.* 2004;1689:91–102.
237. Himmelfarb J, McMonagle E, McMenamin E. Plasma protein thiol oxidation and carbonyl formation in chronic renal failure. *Kidney Int.* 2000;58:2571–2578.
238. Himmelfarb J, McMenamin E, McMonagle E. Plasma aminothiol oxidation in chronic hemodialysis patients. *Kidney Int.* 2002;61:705–716.
239. Miyata T, Sugiyama S, Saito A, Kurokawa K. Reactive carbonyl compounds related uremic toxicity ("carbonyl stress"). *Kidney Int Suppl.* 2001;78:S25–S31.
240. Schwedler SB, Metzger T, Schinzel R, Wanner C. Advanced glycation end products and mortality in hemodialysis patients. *Kidney Int.* 2002;62:301–310.
241. Thornalley PJ, Rabbani N. Highlights and hotspots of protein glycation in end-stage renal disease. *Semin Dial.* 2009;22:400–404.
242. Fumeron C, Nguyen-Khoa T, Saltiel C, et al. Effects of oral vitamin C supplementation on oxidative stress and inflammation status in haemodialysis patients. *Nephrol Dial Transplant.* 2005;20:1874–1879.
243. Lu L, Erhard P, Salomon RG, Weiss MF. Serum vitamin E and oxidative protein modification in hemodialysis: a randomized clinical trial. *Am J Kidney Dis.* 2007;50:305–313.
244. Apeland T, Kristensen O, Mansoor MA. The aminothiol redox status in haemodialysis patients does not improve with folate therapy. *Scand J Clin Lab Invest.* 2009;69:265–271.
245. Ramos LF, Kane J, McMonagle E, et al. Effects of combination tocopherols and alpha lipoic acid therapy on oxidative stress and inflammatory biomarkers in chronic kidney disease. *J Ren Nutr.* 2011;21:211–218.
246. Himmelfarb J, Ikizler TA, Ellis C, et al. Provision of antioxidant therapy in hemodialysis (PATH): a randomized clinical trial. *J Am Soc Nephrol.* 2014;25:623–633.
247. Kaysen GA, Chin A. If oxidative stress is an appropriate and specific target, what reagent should we choose? *J Am Soc Nephrol.* 2014;25:427–429.
248. Monteon FJ, Laidlaw SA, Shaib JK, Kopple JD. Energy expenditure in patients with chronic renal failure. *Kidney Int.* 1986;30:741–747.
249. Veeneman JM, Kingma HA, Stellaard F, de Jong PE, Reijngoud DJ, Huisman RM. Oxidative metabolism appears to be reduced in long-term hemodialysis patients. *Am J Kidney Dis.* 2005;46:102–110.
250. Panesar A, Agarwal R. Resting energy expenditure in chronic kidney disease: relationship with glomerular filtration rate. *Clin Nephrol.* 2003;59:360–366.
251. Ikizler TA, Wingard RL, Sun M, Harvell J, Parker RA, Hakim RM. Increased energy expenditure in hemodialysis patients. *J Am Soc Nephrol.* 1996;7:2646–2653.

252. Avesani CM, Draibe SA, Kamimura MA, Dalboni MA, Colugnati FA, Cuppari L. Decreased resting energy expenditure in non-dialysed chronic kidney disease patients. *Nephrol Dial Transplant.* 2004;19:3091–3097.
253. Wang AY, Sea MM, Tang N, et al. Resting energy expenditure and subsequent mortality risk in peritoneal dialysis patients. *J Am Soc Nephrol.* 2004;15:3134–3143.
254. Shah A, Bross R, Shapiro BB, Morrison G, Kopple JD. Dietary energy requirements in relatively healthy maintenance hemodialysis patients estimated from long-term metabolic studies. *Am J Clin Nutr.* 2016;103:757–765.
255. Mak RH, Cheung W, Cone RD, Marks DL. Leptin and inflammation-associated cachexia in chronic kidney disease. *Kidney Int.* 2006;69:794–797.
256. Spoto B, Pisano A, Zoccali C. Insulin resistance in chronic kidney disease: a systematic review. *Am J Physiol Ren Physiol.* 2016;311:F1087–F1108.
257. Kobayashi S, Maesato K, Moriya H, Ohtake T, Ikeda T. Insulin resistance in patients with chronic kidney disease. *Am J Kidney Dis.* 2005;45:275–280.
258. Sechi LA, Catena C, Zingaro L, Melis A, De Marchi S. Abnormalities of glucose metabolism in patients with early renal failure. *Diabetes.* 2002;51:1226–1232.
259. Hung AM, Ikizler TA. Factors determining insulin resistance in chronic hemodialysis patients. *Contrib Nephrol.* 2011;171:127–134.
260. Maloff BL, McCaleb ML, Lockwood DH. Cellular basis of insulin resistance in chronic uremia. *Am J Physiol.* 1983;245:E178–E184.
261. Smith D, DeFronzo RA. Insulin resistance in uremia mediated by postbinding defects. *Kidney Int.* 1982;22:54–62.
262. Rigalleau V, Gin H. Carbohydrate metabolism in uraemia. *Curr Opin Clin Nutr Metab Care.* 2005;8:463–469.
263. Axelsson J. The emerging biology of adipose tissue in chronic kidney disease: from fat to facts. *Nephrol Dial Transplant.* 2008;23:3041–3046.
264. Reaich D, Channon SM, Scrimgeour CM, Daley SE, Wilkinson R, Goodship TH. Correction of acidosis in humans with CRF decreases protein degradation and amino acid oxidation. *Am J Physiol.* 1993;265:E230–E235.
265. Chapagain A, Caton PW, Kieswich J, et al. Elevated hepatic 11beta-hydroxysteroid dehydrogenase type 1 induces insulin resistance in uremia. *Proc Natl Acad Sci U S A.* 2014;111:3817–3822.
266. Ziegelmeier M, Bachmann A, Seeger J, et al. Serum levels of adipokine retinol-binding protein-4 in relation to renal function. *Diabetes Care.* 2007;30:2588–2592.
267. Wojtaszewski JF, Jorgensen SB, Frosig C, MacDonald C, Birk JB, Richter EA. Insulin signalling: effects of prior exercise. *Acta Physiol Scand.* 2003;178:321–328.
268. Mustata S, Chan C, Lai V, Miller JA. Impact of an exercise program on arterial stiffness and insulin resistance in hemodialysis patients. *J Am Soc Nephrol.* 2004;15:2713–2718.
269. Michou V, Tsamos G, Vasdeki D, Deligiannis A, Kouidi E. Unraveling of molecular mechanisms of cognitive frailty in chronic kidney disease: how exercise makes a difference. *J Clin Med.* 2024;13.
270. Valenzuela PL, Castillo-García A, Saco-Ledo G, Santos-Lozano A, Lucia A. Physical exercise: a polypill against chronic kidney disease. *Nephrol Dial Transplant.* 2024;39:1384–1391.
271. Shinohara K, Shoji T, Emoto M, et al. Insulin resistance as an independent predictor of cardiovascular mortality in patients with end-stage renal disease. *J Am Soc Nephrol.* 2002;13:1894–1900.
272. DeFronzo RA. Insulin and renal sodium handling: clinical implications. *Int J Obes.* 1981;5(suppl 1):93–104.
273. Brands MW, Hall JE, Keen HL. Is insulin resistance linked to hypertension? *Clin Exp Pharmacol Physiol.* 1998;25:70–76.
274. Lim VS, Kopple JD. Protein metabolism in patients with chronic renal failure: role of uremia and dialysis. *Kidney Int.* 2000;58:1–10.
275. Adrogue HJ. Glucose homeostasis and the kidney. *Kidney Int.* 1992;42:1266–1282.
276. Owen OE, Felig P, Morgan AP, Wahren J, Cahill Jr GF. Liver and kidney metabolism during prolonged starvation. *J Clin Investig.* 1969;48:574–583.
277. Cersosimo E, Judd RL, Miles JM. Insulin regulation of renal glucose metabolism in conscious dogs. *J Clin Investig.* 1994;93:2584–2589.
278. Sahadevan M, Kasiske BL. Hyperlipidemia in kidney disease: causes and consequences. *Curr Opin Nephrol Hypertens.* 2002;11:323–329.
279. Kwan BC, Kronenberg F, Beddhu S, Cheung AK. Lipoprotein metabolism and lipid management in chronic kidney disease. *J Am Soc Nephrol.* 2007;18:1246–1261.
280. Kronenberg F, Ikewaki K, Schaefer JR, Konig P, Dieplinger H. Kinetic studies of atherogenic lipoproteins in hemodialysis patients: do they tell us more about their pathology? *Semin Dial.* 2007;20:554–560.
281. Baigent C, Landray MJ, Reith C, et al. The effects of lowering LDL cholesterol with simvastatin plus ezetimibe in patients with chronic kidney disease (Study of Heart and Renal Protection): a randomised placebo-controlled trial. *Lancet.* 2011;377:2181–2192.
282. Wanner C, Krane V, Marz W, et al. Atorvastatin in patients with type 2 diabetes mellitus undergoing hemodialysis. *N Engl J Med.* 2005;353:238–248.
283. Fellstrom BC, Jardine AG, Schmieder RE, et al. Rosuvastatin and cardiovascular events in patients undergoing hemodialysis. *N Engl J Med.* 2009;360:1395–1407.
284. Palmer SC, Navaneethan SD, Craig JC, et al. HMG CoA reductase inhibitors (statins) for dialysis patients. *Cochrane Database Syst Rev.* 2013:CD004289.
285. Su X, Zhang L, Lv J, et al. Effect of statins on kidney disease outcomes: a systematic review and meta-analysis. *Am J Kidney Dis.* 2016;67:881–892.
286. Palmer SC, Craig JC, Navaneethan SD, Tonelli M, Pellegrini F, Strippoli GF. Benefits and harms of statin therapy for persons with chronic kidney disease: a systematic review and meta-analysis. *Ann Intern Med.* 2012;157:263–275.
287. Sarnak MJ, Bloom R, Muntner P, et al. KDOQI US commentary on the 2013 KDIGO clinical practice guideline for lipid management in CKD. *Am J Kidney Dis.* 2015;65:354–366.
288. Kopple JD, Jones M, Fukuda S, Swendseid ME. Amino acid and protein metabolism in renal failure. *Am J Clin Nutr.* 1978;31:1532–1540.
289. Laidlaw SA, Berg RL, Kopple JD, Naito H, Walker WG, Walser M. Patterns of fasting plasma amino acid levels in chronic renal insufficiency: results from the feasibility phase of the Modification of Diet in Renal Disease Study. *Am J Kidney Dis.* 1994;23:504–513.
290. Ceballos I, Chauveau P, Guerin V, et al. Early alterations of plasma free amino acids in chronic renal failure. *Clin Chim Acta.* 1990;188:101–108.
291. Tizianello A, De Ferrari G, Garibotto G, Gurreri G, Robaudo C. Renal metabolism of amino acids and ammonia in subjects with normal renal function and in patients with chronic renal insufficiency. *J Clin Investig.* 1980;65:1162–1173.
292. Francis ME, Eggers PW, Hostetter TH, Briggs JP. Association between serum homocysteine and markers of impaired kidney function in adults in the United States. *Kidney Int.* 2004;66:303–312.
293. Jamison RL, Hartigan P, Kaufman JS, et al. Effect of homocysteine lowering on mortality and vascular disease in advanced chronic kidney disease and end-stage renal disease: a randomized controlled trial. *JAMA.* 2007;298:1163–1170.
294. Rajan VR, Mitch WE. Muscle wasting in chronic kidney disease: the role of the ubiquitin proteasome system and its clinical impact. *Pediatr Nephrol.* 2008;23:527–535.
295. Garibotto G, Sofia A, Russo R, et al. Insulin sensitivity of muscle protein metabolism is altered in patients with chronic kidney disease and metabolic acidosis. *Kidney Int.* 2015;88:1419–1426.
296. Graham KA, Reaich D, Channon SM, et al. Correction of acidosis in CAPD decreases whole body protein degradation. *Kidney Int.* 1996;49:1396–1400.
297. Graham KA, Reaich D, Channon SM, Downie S, Goodship TH. Correction of acidosis in hemodialysis decreases whole-body protein degradation. *J Am Soc Nephrol.* 1997;8:632–637.
298. Lofberg E, Gutierrez A, Anderstam B, et al. Effect of bicarbonate on muscle protein in patients receiving hemodialysis. *Am J Kidney Dis.* 2006;48:419–429.
299. Levin A, Ahmed SB, Carrero JJ, et al. Executive summary of the KDIGO 2024 clinical practice guideline for the evaluation and management of chronic kidney disease: known knowns and known unknowns. *Kidney Int.* 2024;105:684–701.
300. Kadatane SP, Satariano M, Massey M, Mongan K, Raina R. The role of inflammation in CKD. *Cells.* 2023;12.
301. Goldstein SL, Ikizler TA, Zappitelli M, Silverstein DM, Ayus JC. Non-infected hemodialysis catheters are associated with increased inflammation compared to arteriovenous fistulas. *Kidney Int.* 2009;76:1063–1069.

302. Ledebo I. Ultrapure dialysis fluid–how pure is it and do we need it? *Nephrol Dial Transplant.* 2007;22:20–23.
303. Stenvinkel P, Carrero JJ, Axelsson J, Lindholm B, Heimburger O, Massy Z. Emerging biomarkers for evaluating cardiovascular risk in the chronic kidney disease patient: how do new pieces fit into the uremic puzzle? *Clin J Am Soc Nephrol.* 2008;3:505–521.
304. Rivara MB, Yeung CK, Robinson-Cohen C, et al. Effect of coenzyme Q(10) on biomarkers of oxidative stress and cardiac function in hemodialysis patients: the CoQ(10) biomarker trial. *Am J Kidney Dis.* 2017;69:389–399.
305. Nowak KL, Chonchol M, Ikizler TA, et al. IL-1 inhibition and vascular function in CKD. *J Am Soc Nephrol.* 2017;28:971–980.
306. Thornalley PJ. Advanced glycation end products in renal failure. *J Ren Nutr.* 2006;16:178–184.
307. Galli F. Protein damage and inflammation in uraemia and dialysis patients. *Nephrol Dial Transplant.* 2007;22(Suppl 5):v20–v36.
308. Johansen KL, Chertow GM, Kutner NG, Dalrymple LS, Grimes BA, Kaysen GA. Low level of self-reported physical activity in ambulatory patients new to dialysis. *Kidney Int.* 2010;78:1164–1170.
309. Johansen KL, Kaysen GA, Dalrymple LS, et al. Association of physical activity with survival among ambulatory patients on dialysis: the Comprehensive Dialysis Study. *Clin J Am Soc Nephrol.* 2013;8:248–253.
310. Anand S, Johansen KL, Grimes B, et al. Physical activity and self-reported symptoms of insomnia, restless legs syndrome, and depression: the comprehensive dialysis study. *Hemodial Int.* 2013;17:50–58.
311. Wang K, Liu Q, Tang M, et al. Chronic kidney disease-induced muscle atrophy: molecular mechanisms and promising therapies. *Biochem Pharmacol.* 2023;208:115407.
312. Viana JL, Kosmadakis GC, Watson EL, et al. Evidence for anti-inflammatory effects of exercise in CKD. *J Am Soc Nephrol.* 2014;25:2121–2130.
313. Watson EL, Baker LA, Wilkinson TJ, et al. Inflammation and physical dysfunction: responses to moderate intensity exercise in chronic kidney disease. *Nephrol Dial Transplant.* 2022;37:860–868.
314. Kaysen GA. Association between inflammation and malnutrition as risk factors of cardiovascular disease. *Blood Purif.* 2006;24:51–55.
315. Wulczyn K, Zhao S, Rhee E, Kalim S, Shafi T. Trajectories of uremic symptom severity and kidney function in patients with chronic kidney disease. *Clin J Am Soc Nephrol.* 2022;17:496–506.
316. Wulczyn KE, Zhao SH, Rhee EP, Kalim S, Shafi T. Trajectories of uremic symptom severity and kidney function in patients with chronic kidney disease. *Clin J Am Soc Nephrol.* 2022;17:496–506.
317. Ring T, Merkus MP, Krediet RT. Physical symptoms and quality of life in patients on chronic dialysis: results of The Netherlands co-operative study on adequacy of dialysis (NECOSAD). *Nephrol Dial Transplant.* 2000;15:280–281.
318. Kimmel PL, Emont SL, Newmann JM, Danko H, Moss AH. ESRD patient quality of life: symptoms, spiritual beliefs, psychosocial factors, and ethnicity. *Am J Kidney Dis.* 2003;42:713–721.
319. Weisbord SD, Fried LF, Arnold RM, et al. Prevalence, severity, and importance of physical and emotional symptoms in chronic hemodialysis patients. *J Am Soc Nephrol.* 2005;16:2487–2494.
320. Perlman RL, Finkelstein FO, Liu L, et al. Quality of life in chronic kidney disease (CKD): a cross-sectional analysis in the Renal Research Institute-CKD study. *Am J Kidney Dis.* 2005;45:658–666.
321. Walters BA, Hays RD, Spritzer KL, Fridman M, Carter WB. Health-related quality of life, depressive symptoms, anemia, and malnutrition at hemodialysis initiation. *Am J Kidney Dis.* 2002;40:1185–1194.
322. Wu AW, Fink NE, Marsh-Manzi JV, et al. Changes in quality of life during hemodialysis and peritoneal dialysis treatment: generic and disease specific measures. *J Am Soc Nephrol.* 2004;15:743–753.
323. Abdel-Kader K, Unruh ML, Weisbord SD. Symptom burden, depression, and quality of life in chronic and end-stage kidney disease. *Clin J Am Soc Nephrol.* 2009;4:1057–1064.
324. Mehrotra R, Davison SN, Farrington K, et al. Managing the symptom burden associated with maintenance dialysis: conclusions from a kidney disease: improving global outcomes (KDIGO) controversies conference. *Kidney Int.* 2023;104:441–454.
325. de Rooij ENM, Meuleman Y, de Fijter JW, et al. Symptom burden before and after dialysis initiation in older patients. *Clin J Am Soc Nephrol.* 2022;17:1719–1729.
326. de Rooij ENM, Meuleman Y, de Fijter JW, et al. Quality of life before and after the start of dialysis in older patients. *Clin J Am Soc Nephrol.* 2022;17:1159–1167.
327. Chiu YW, Teitelbaum I, Misra M, de Leon EM, Adzize T, Mehrotra R. Pill burden, adherence, hyperphosphatemia, and quality of life in maintenance dialysis patients. *Clin J Am Soc Nephrol.* 2009;4:1089–1096.
328. Kimmel PL, Patel SS. Quality of life in patients with chronic kidney disease: focus on end-stage renal disease treated with hemodialysis. *Semin Nephrol.* 2006;26:68–79.
329. Myint TM, Cooper BA, Pollock CA, Harris DC. Starting dialysis early: no survival, quality of life, or cost advantages. *Semin Dial.* 2012;25:522–523.
330. Roshanravan B, Gamboa J, Wilund K. Exercise and CKD: skeletal muscle dysfunction and practical application of exercise to prevent and treat physical impairments in CKD. *Am J Kidney Dis.* 2017;69:837–852.
331. Johansen KL. Physical functioning and exercise capacity in patients on dialysis. *Adv Ren Replace Ther.* 1999;6:141–148.
332. Painter P. Physical functioning in end-stage renal disease patients: update 2005. *Hemodial Int.* 2005;9:218–235.
333. Johansen KL, Doyle J, Sakkas GK, Kent-Braun JA. Neural and metabolic mechanisms of excessive muscle fatigue in maintenance hemodialysis patients. *Am J Physiol Regul Integr Comp Physiol.* 2005;289:R805–R813.
334. Blake C, O'Meara YM. Subjective and objective physical limitations in high-functioning renal dialysis patients. *Nephrol Dial Transplant.* 2004;19:3124–3129.
335. Moorthi RN, Fadel WF, Cranor A, et al. Mobility impairment in patients new to dialysis. *Am J Nephrol.* 2020;51:705–714.
336. Brouns R, De Deyn PP. Neurological complications in renal failure: a review. *Clin Neurol Neurosurg.* 2004;107:1–16.
337. O'Lone E, Connors M, Masson P, et al. Cognition in people with end-stage kidney disease treated with hemodialysis: a systematic review and meta-analysis. *Am J Kidney Dis.* 2016;67:925–935.
338. Murray AM, Pederson SL, Tupper DE, et al. Acute variation in cognitive function in hemodialysis patients: a cohort study with repeated measures. *Am J Kidney Dis.* 2007;50:270–278.
339. Kurella M, Chertow GM, Fried LF, et al. Chronic kidney disease and cognitive impairment in the elderly: the health, aging, and body composition study. *J Am Soc Nephrol.* 2005;16:2127–2133.
340. Kurella M, Yaffe K, Shlipak MG, Wenger NK, Chertow GM. Chronic kidney disease and cognitive impairment in menopausal women. *Am J Kidney Dis.* 2005;45:66–76.
341. Hailpern SM, Melamed ML, Cohen HW, Hostetter TH. Moderate chronic kidney disease and cognitive function in adults 20 to 59 years of age: third National Health and Nutrition Examination Survey (NHANES III). *J Am Soc Nephrol.* 2007;18:2205–2213.
342. Madero M, Gul A, Sarnak MJ. Cognitive function in chronic kidney disease. *Semin Dial.* 2008;21:29–37.
343. Seliger SL, Longstreth Jr WT, Katz R, et al. Cystatin C and subclinical brain infarction. *J Am Soc Nephrol.* 2005;16:3721–3727.
344. Kim CD, Lee HJ, Kim DJ, et al. High prevalence of leukoaraiosis in cerebral magnetic resonance images of patients on peritoneal dialysis. *Am J Kidney Dis.* 2007;50:98–107.
345. Griva K, Thompson D, Jayasena D, Davenport A, Harrison M, Newman SP. Cognitive functioning pre- to post-kidney transplantation--a prospective study. *Nephrol Dial Transplant.* 2006;21:3275–3282.
346. Harciarek M, Biedunkiewicz B, Lichodziejewska-Niemierko M, Debska-Slizien A, Rutkowski B. Continuous cognitive improvement 1 year following successful kidney transplant. *Kidney Int.* 2011;79:1353–1360.
347. Van Sandwijk MS, Ten Berge IJ, Majoie CB, et al. Cognitive changes in chronic kidney disease and after transplantation. *Transplantation.* 2016;100:734–742.
348. Schneider SM, Malecki AK, Muller K, et al. Effect of a single dialysis session on cognitive function in CKD5D patients: a prospective clinical study. *Nephrol Dial Transplant.* 2015;30:1551–1559.
349. Perl J, Unruh ML, Chan CT. Sleep disorders in end-stage renal disease: 'Markers of inadequate dialysis. *Kidney Int.* 2006;70:1687–1693.
350. Unruh ML, Larive B, Eggers PW, et al. The effect of frequent hemodialysis on self-reported sleep quality: Frequent Hemodialysis Network Trials. *Nephrol Dial Transplant.* 2016;31:984–991.

351. Mucsi I, Molnar MZ, Ambrus C, et al. Restless legs syndrome, insomnia and quality of life in patients on maintenance dialysis. *Nephrol Dial Transplant*. 2005;20:571–577.
352. Gopaluni S, Sherif M, Ahmadouk NA. Interventions for chronic kidney disease-associated restless legs syndrome. *Cochrane Database Syst Rev*. 2016;11:CD010690.
353. Krishnan AV, Phoon RK, Pussell BA, Charlesworth JA, Bostock H, Kiernan MC. Altered motor nerve excitability in end-stage kidney disease. *Brain*. 2005;128:2164–2174.
354. Krishnan AV, Kiernan MC. Uremic neuropathy: clinical features and new pathophysiological insights. *Muscle Nerve*. 2007;35:273–290.
355. Sukul N, Speyer E, Tu C, et al. Pruritus and patient reported outcomes in non-dialysis CKD. *Clin J Am Soc Nephrol*. 2019;14:673–681.
356. Mathur VS, Lindberg J, Germain M, et al. A longitudinal study of uremic pruritus in hemodialysis patients. *Clin J Am Soc Nephrol*. 2010;5:1410–1419.
357. Sukul N, Karaboyas A, Csomor PA, et al. Self-reported pruritus and clinical, dialysis-related, and patient-reported outcomes in hemodialysis patients. *Kidney Med*. 2021;3:42–53.
358. Davison SN, Levin A, Moss AH, et al. Executive summary of the KDIGO controversies conference on supportive care in chronic kidney disease: developing a roadmap to improving quality care. *Kidney Int*. 2015;88:447–459.
359. Fishbane S, Jamal A, Munera C, Wen W, Menzaghi F. A phase 3 trial of difelikefalin in hemodialysis patients with pruritus. *N Engl J Med*. 2020;382:222–232.
360. Rayner HC, Larkina M, Wang M, et al. International comparisons of prevalence, awareness, and treatment of pruritus in people on hemodialysis. *Clin J Am Soc Nephrol*. 2017;12:2000–2007.
361. Wulczyn KE, Rhee EP, Myint L, Kalim S, Shafi T. Incidence and risk factors for pruritus in patients with nondialysis CKD. *Clin J Am Soc Nephrol*. 2023;18:193–203.
362. Makar M, Smyth B, Brennan F. Chronic kidney disease-associated pruritus: a review. *Kidney Blood Press Res*. 2021;46:659–669.
363. Bergstrom J. Regulation of appetite in chronic renal failure. *Miner Electrolyte Metab*. 1999;25:291–297.
364. Mitch WE. Cachexia in chronic kidney disease: a link to defective central nervous system control of appetite. *J Clin Investig*. 2005;115:1476–1478.
365. Cheung W, Yu PX, Little BM, Cone RD, Marks DL, Mak RH. Role of leptin and melanocortin signaling in uremia-associated cachexia. *J Clin Investig*. 2005;115:1659–1665.
366. Cheung WW, Ding W, Gunta SS, et al. A pegylated leptin antagonist ameliorates CKD-associated cachexia in mice. *J Am Soc Nephrol*. 2014;25:119–128.
367. Griep MI, Van der Niepen P, Sennesael JJ, Mets TF, Massart DL, Verbeelen DL. Odour perception in chronic renal disease. *Nephrol Dial Transplant*. 1997;12:2093–2098.
368. Middleton RA, Allman-Farinelli MA. Taste sensitivity is altered in patients with chronic renal failure receiving continuous ambulatory peritoneal dialysis. *J Nutr*. 1999;129:122–125.
369. Chewcharat A, Phipps EA, Bhatia K, et al. The association between olfactory and gustatory dysfunction and chronic kidney disease. *BMC Nephrol*. 2022;23:36.
370. Nigwekar SU, Weiser JM, Kalim S, et al. Characterization and correction of olfactory deficits in kidney disease. *J Am Soc Nephrol*. 2017;28:3395–3403.
371. Fernstrom A, Hylander B, Rossner S. Taste acuity in patients with chronic renal failure. *Clin Nephrol*. 1996;45:169–174.
372. Lynch KE, Lynch R, Curhan GC, Brunelli SM. Altered taste perception and nutritional status among hemodialysis patients. *J Ren Nutr*. 2013;23:288–295.e1.
373. Welt LG, Sachs JR, McManus TJ. An ion transport defect in erythrocytes from uremic patients. *Trans Assoc Am Phys*. 1964;77:169–181.
374. Kaji D, Thomas K. Na+-K+ pump in chronic renal failure. *Am J Physiol*. 1987;252:F785–F793.
375. Singer MA, Morton AR. Mouse to elephant: biological scaling and Kt/V. *Am J Kidney Dis*. 2000;35:306–309.
376. Beuchat CA. Structure and concentrating ability of the mammalian kidney: correlations with habitat. *Am J Physiol*. 1996;271:R157–R179.
377. Nelson RA. Black bears and polar bears–still metabolic marvels. *Mayo Clin Proc*. 1987;62:850–853.
378. Stenvinkel P, Jani AH, Johnson RJ. Hibernating bears (Ursidae): metabolic magicians of definite interest for the nephrologist. *Kidney Int*. 2013;83:207–212.
379. Ohtsuki S. New aspects of the blood-brain barrier transporters; its physiological roles in the central nervous system. *Biol Pharm Bull*. 2004;27:1489–1496.
380. Koeth RA, Kalantar-Zadeh K, Wang Z, Fu X, Tang WH, Hazen SL. Protein carbamylation predicts mortality in ESRD. *J Am Soc Nephrol*. 2013;24:853–861.
381. Tang WH, Shrestha K, Wang Z, et al. Protein carbamylation in chronic systolic heart failure: relationship with renal impairment and adverse long-term outcomes. *J Card Fail*. 2013;19:219–224.
382. Kalim S, Tamez H, Wenger J, et al. Carbamylation of serum albumin and erythropoietin resistance in end stage kidney disease. *Clin J Am Soc Nephrol*. 2013;8:1927–1934.
383. Drechsler C, Kalim S, Wenger JB, et al. Protein carbamylation is associated with heart failure and mortality in diabetic patients with end-stage renal disease. *Kidney Int*. 2015;87:1201–1208.
384. Kalim S, Berg AH, Karumanchi SA, et al. Protein carbamylation and chronic kidney disease progression in the chronic renal insufficiency cohort study. *Nephrol Dial Transplant*. 2021;37:139–147.
385. Tan KCB, Cheung CL, Lee ACH, Lam JKY, Wong Y, Shiu SWM. Carbamylated lipoproteins and progression of diabetic kidney disease. *Clin J Am Soc Nephrol*. 2020;15:359–366.